英语科技写作

［第二版］

Hilary Glasman-Deal

［英］希拉里·格拉斯曼-蒂欧 ——— 著

雷锦志　李晓文　武林晓 ——— 译

Science Research Writing

Second Edition

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这本书应该出现在每位STEMM学者的书架上。

克里斯托弗·杰克逊，帝国理工学院教授

本书不仅在通常来看较为晦涩的领域为新手写作者提供了非常有用的方法，还给出了深入的分析和实际的论证，这将使经验丰富的STEMM研究人员大放异彩。

辛西娅·怀特，新西兰梅西大学副校长

作为一名科学研究员，你能够无障碍地阅读和理解你所在领域的高度复杂的材料。然而，你发现自己却很难撰写一篇与你阅读水平相当的论文，或者你可能会感觉到你的文章不能有效地传达研究的内容和影响力。在本书中，你将学习如何利用现有的阅读能力和所阅读的材料，提升相关的写作技巧。你不需要学习更多的知识，只需要有效运用自己已经掌握的知识，而不是想到哪写到哪。科学写作要比看起来的更容易。

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第二版译者序

很高兴能够翻译本书的第二版。从第一版开始，本书就一直是我和我的学生在科技写作方面不可缺少的参考书。近10年来，因为网络发表和获取学术论文的方式发生变化，科技写作也经历了很大的变化，很多期刊也因此对论文的形式和内容都提出了许多新的要求。第二版针对这些新情况进行了比较大的调整。更重要的一点是，在第二版的书中，作者不再强调一种一成不变的写作模板，而是教会读者方法，通过分析已经发表的论文总结出适合自己的写作模板。可以说，这是从“授之以鱼”到“授之以渔”的跨越。

在阅读本书的过程中，我很深刻地感受到了科技写作过程中作者与读者有效沟通的重要性。有很多内容在作者心里是明白的，但是读者不一定明白。有时作者自己对内容的熟悉程度

反而会是他成为好的科技写作者的障碍。通过阅读这本书，我很强烈地感受到，科技写作的目的不仅仅是使读者能够理解，而且是要使读者不可能理解不了。这一点应该成为好的科技写作的标准。

本书的写作延续了第一版结构清晰、叙述简练易懂、重点突出的风格。每一章都通过大量的实例，一步一步地教读者如何从目标论文中总结出实用的写作模板，同时结合写作内容重点阐述相关的语法问题。

应该要重点关注本书第二版的两个重要变化。第一个变化是如前面所述的，主要目的不是建立一成不变的模板，而是帮助读者建立日后不会被淘汰的写作策略。第二个变化是本书面向的读者群。第一版的读者对象是非英语母语的科技作者。而在第二版中，作者强调科技写作是所有科技工作者共同的问题，“学术英文并非每个人的母语”。特别是在语法和内容方面，使用正确的语法并不能保证科技文章表述的准确性，相反，一篇在语言表达上有误但写得流畅并能准确地和读者交流的作品却能够成功发表。因此，在科技写作方面，首先要知道应该写什么内容，然后是如何用正确的语法表达这些内容。

本书最后一单元是清单和提示，可以看作是对整本书的内容的总结。对于想尽快开始科技写作的年轻学者和学生来说，也许首先阅读这一单元是有帮助的。

作为研究生导师，我深知论文写作是每一位研究生必须要迈过的一道坎。我向大家推荐本书，希望它能帮助大家尽快地

进入科技写作的行列。

最后，译文难免有错误和不妥之处，恳请读者批评指正。

雷锦志

天津工业大学

第一版译者序

第一次看到这本书是2009年我在新加坡南洋理工大学高等研究院访问期间。那是在参加一次小组讨论前，黄克孙教授特意把这本书交给我，说："这本书很好，你要好好读读！"黄先生是美国麻省理工学院的物理学退休教授，目前受聘为南洋理工大学高等研究院的资深教授。我和黄先生近年来一直合作从事蛋白质折叠机理方面的研究。我想既然黄先生说是好书，那一定是好的。就这样，我开始认真阅读，并很快为之所吸引，深感其内容对科技论文的写作有很大帮助。于是就萌生了要翻译本书以推荐给国内科技工作者的想法。

这本书结构清晰、叙述简练易懂、重点突出，并且给出了很多优秀的科技论文写作范例，是一本非常实用的参考书。

本书具有以下特点：

- 内容安排合理。全书按照论文写作的顺序进行布局，分别是引言、方法、结果、讨论/结论、摘要。这种安排方便读者在写作的时候查阅所需的资料。
- 每部分的结构相似，都是从实例出发，一步一步地教读者根据实例总结出该部分的写作模板，同时介绍讨论写这一部分时所需的语法知识和写作技巧。这些模板、语法知识和写作技巧在具体写论文的时候都非常有用。
- 每部分都列举出很多实用的单词，并且按照这些单词的用法和内容等进行分类。因此，本书可以作为很好的手册供科技工作者参考。
- 对语法的介绍并非大而全（这样的内容在大部分外语教科书上都可以看到），而是重点讨论那些非英语母语人士容易犯的错误，例如时态的运用、a 和 the 的使用等。针对这些内容，作者用非常简洁而清晰的例子告诉读者如何正确地表达自己想表达的意思，这对广大科技工作者写英语论文是非常有帮助的。
- 本书详细讨论了学术论文中每个部分应该写什么和如何写，这些内容对于写中文科技论文也是非常有帮助的。

由于译者水平有限，译文难免有错误和不妥之处，恳请读者批评指正。

雷锦志
清华大学，周培源应用数学研究中心
2011年1月

CONTENTS

目 录

第二版前言 001

引 言 007

第一单元

怎么写引言 023

1.1 引言的结构和内容 025

1.2 创建模板 026

1.3 验证和调整模板 044

1.4 有用的词汇和短语 085

1.5 语言和写作技巧 095

1.6 撰写引言 117

第二单元
怎么写方法 123

2.1 方法部分的结构和内容 125
2.2 创建模板 127
2.3 验证和调整模板 143
2.4 有用的词汇和短语 158
2.5 语言和写作技巧 177
2.6 编写方法部分 198

第三单元
怎么写结果 215

3.1 结果的结构和内容 217
3.2 创建模板 218
3.3 验证和调整模板 236
3.4 有用的词汇和短语 256
3.5 语言和写作技巧：确定性连续统 275

第四单元
怎么写讨论 281

4.1 怎么写讨论 283
4.2 建立你自己的模板 284
4.3 验证和调整基本通用模板 305
4.4 有用的词汇和短语 324
4.5 语言和写作技巧：情态动词 335
4.6 概述讨论部分练习 352

第五单元
怎么写结论 355

5.1 创建模板 358
5.2 测试和调整模板 375
5.3 有用的词汇和短语 376
5.4 语言和写作技巧 376

第六单元
怎么写摘要 383

6.1 摘要指南 386
6.2 摘要的类型 390
6.3 通用摘要模板 415
6.4 语言 421
6.5 摘要总结练习 431

第七单元
撰写标题 433

7.1 检查平均长度 438
7.2 使用缩写 439
7.3 将标题关键字与关键字列表进行比较 439
7.4 检查标题的语法 441
7.5 映射和建立目标文章标题的结构性内容的模板 443
7.6 检查标题所建议的期望是否在论文中得到满足 444

第八单元
清单和提示 449

8.1 组织信息 452
8.2 创建句子 456
8.3 语法和词汇 459
8.4 常规 470

致 谢 473
参考资料 475
摘录索引 477
附 录 A：科技写作中的前缀 481
附 录 B：研究中的动词 485

第二版前言

学生：爱因斯坦博士，这些问题和去年提出的不是一样的吗？

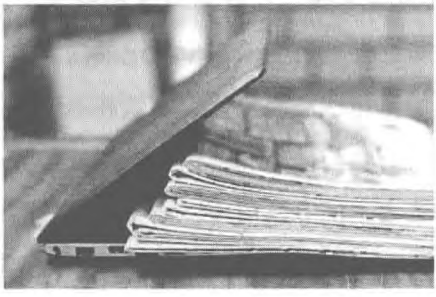
爱因斯坦博士：是的，但今年的答案会不一样。

受到推动科学沟通方式和研究成果获取方式的因素的影响，科学研究写作正在经历日新月异的变化和发展。这些因素包括科学研究的全球性、互联网阅读和科学发现的速度的增加，它们正在革命性地改变着二十世纪的沟通模式。

科学研究的全球性需要快速、有效和准确的沟通方式，以便于世界各地所有参与研究和合作的工作者都能理解。科学语言不是英语母语人士的专属语言，而是全球性的语言。非英语母语的学者经常在自己的科学领域内遥遥领先。这些学者在高

影响力的期刊上汇报他们的工作，并需要快速而有效地撰写和阅读他们学科的研究成果。这些变化正在创造一种全球化的科学沟通语言，使不同国家、不同层次的所有读者都能使用它进行科研写作。

互联网阅读是另一个正在改变研究成果获取方式的因素，同时也影响着科研写作。由于 STEMM（科学、技术、工程、数学和医学）研究领域的协作性以及研究经费增加，每年发表的科研论文的数量日益增长，因此，成功的 STEMM 研究所需的阅读量也随之增长了。以往的学者只能阅览一套有限的、经过精心策划的纸质期刊复印件。然而，如今的大多数学者在互联网上通过信息冲浪能更快更流畅地获取更大范围的研究论



文。进行一项研究活动要搜索和研究大量的相关题目的资料，因此读者在搜索的过程中可能只阅读研究论文的一部分，可能是摘要、结果或结论。这种需求的结果是论文的题目变得更重要，并且研究论文的某些部分，特别是摘要或概要，正在演变成可以独立阅读的文本。

第三个推动科学研究沟通方式变化的因素是速度的需要。科学向前发展的速度几乎要快于它能够被报告出来的速度，STEMM 的研究需要作者将极其复杂的信息尽可能快速地传达出来。在大多数情况下，这么做是为了避免被其他研究人员超越，也可能是因为科学研究与行业、专利或者衍生公司之间的某些联系需要它尽快被发表出来。如果论文撰写的时间过长，那么它到了提交的时候就会有部分或全部过期的风险，导致成果不能发表。因此，科学家和刊物编辑需要尽可能快速而简明地传达出一项研究的影响，这也影响着 STEMM 的科研写作。

研究本身的性质也在不断改变：科学在某些方面变得更专门化，而在其他方面变得更跨学科。这造就了复杂的沟通结构，非专业人员有时却需要获取高度专业的信息。

上述变化正深刻地影响传统科研写作的结构和内容。研究论文的形式正在转变，以满足研究界快速变化的需要。内容和摘要的功能正在转变，研究方法这部分的位置也在移动。众所周知的 IMRaD（引言、方法、结果、讨论）结构正在渐渐松散，论文尤其是在线发表的研究论文里同时包含了一些重点标记，如研究背景（Research in Context）部分、长达一页的概

要（Summaries）和研究亮点（Highlights）列表。为了能成功地全面参与到目前的科研环境，科研人员需要意识到科学研究写作的变化和发展，并对此做出回应。他们也应该对将来的改变未雨绸缪。因而，此书第二版做了两个关键的变化。

第二版的两个变化

第一个变化体现了科学沟通平台正在改变的事实。第一版分析了在不同学科有重大影响的研究论文，为研究文章的每个部分生成了一套相当稳定的写作模板。根据目前科研写作趋势的改变，这些写作模式在第二版有了显著的更新，而它们也不再被看作是一成不变的表达形式。随着下一轮的科学沟通方式的变化，研究人员需要培养一种能力以不断地更新自己的写作模式去适应这些变化。鉴于上述的写作模式快速变化的特性，第二版的目的是帮助作者建立面向未来的写作策略，这些策略是一种终身可用的工具，而不管将来科学沟通方式如何变幻莫测。

第二版通过鼓励大家使用逆向工程的方式来培养这些写作策略。大多数阅读都包含从文本中被动地获取信息源，而对文本的逆向工程与之相反，它需要用主动的方式去阅读，包括学会从信息本身后退一步，从而留意信息的传递方式。一旦掌握了这样的策略，STEMM 科研工作者便可以利用逆向工程作为写作的起点，在现在和将来的不同学术交流场合撰写不同类型的文本，包括计划书、会议摘要、书信、科技报告和评论报告。

逆向工程的方式和数字绘画有异曲同工之妙。数字绘画的营销口号是“第一次尝试你就能成功创造一幅油画佳作”，使用这种数字绘画的方式也许不能让你成为达·芬奇，但是它可以传递一种策略，该策略能帮你创造出令人难以辨认的相似画作。

https://en.wikipedia.org/wiki/Mona_lisa

同样地，分析一组写得很棒的研究论文，它的构成、写作模式和语言的运用会启发我们形成一种策略，这种策略会帮助我们撰写一篇适合当前惯例、有可信度、井井有条、可读性高的研究论文。

第二版的第二个变化是本书面向的读者群。第一版的阅读对象是非英语母语的科技作者。然而我和非英语母语的科技人员多年的合作经验清楚地告诉我，虽然非英语母语的科技作者

有特定的语言问题，但写作问题是所有 STEMM 作者共同的问题。所有科研人员写作的目的都是根据大量的数据和信息写出成功和有效的文字。STEMM 科研项目的复杂性需要作者有相当的沟通技能。然而，“学术英文并非每个人的母语”[1]，即便是最高水平的英语母语研究者，他们在撰写科研文章时也要面对和非英语母语者同样的写作问题。从我的经验来看，使用正确的英语语法并不能保证一篇科技文章就是好作品；相反，尽管在语言表达上有误，一篇写得流畅并能准确地和读者交流的作品就能登上成功文章的宝座。因此，我将本书的重点转移到 STEMM 学者共同面对的语言和写作问题，特别是那些影响到文章的表达和可读性的问题，不管学者们的母语是不是英语。

自从这本书第一版在2009年问世之后，《英语科技写作》（*Science Research Writing*）已被译成中文、日文和韩文，并被全球多个大学和科研机构以及它们的讲师、学生和科研人员采用，包括来自学术英语（English for Academic Purpose, EAP）与专业英语（English for Specific Academic Purposes, ESAP）的讲师和学生，以及英语母语的 STEMM 科研人员。这些读者为第二版的改编提供了很多帮助，包括添加清单和提示章节，这些章节对其他单元所处理的写作问题进行了总结。

引言

为读者写作：将信息包装在叙述中。

科学研究写作不仅仅是提供信息，哪怕它表达得再清晰。首先，信息本身是中性的——它本身不具有内在的功能。如果作者不直截了当地指明和传达信息的功能以及这些信息与文章其他部分的关系，那么读者就不会明白这些信息为什么会在文章中出现。这样就会打断文章的连贯性从而影响其可读性。其次，科学研究写作的目的是增加或者改变读者的知识，这需要围绕信息的强有力的叙述手法，有逻辑和有耐心地推断出作者希望得出的结论。

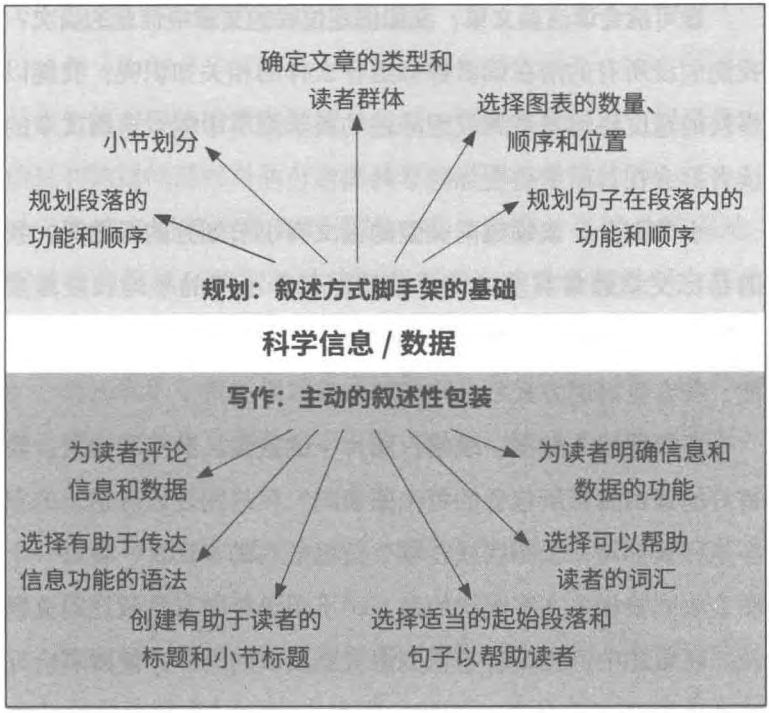
作者对自己研究内容的高度熟悉使他们很难理解为什么好的叙述很必要。在开始写作以前，作者可能和几个同事在数月前就保持着紧密合作并分享关于研究主题和研究内容的一般性和特定的知识。组内的交流通常是不言自明的，因为那些同

事都知道为什么（why）、怎么做（how）和做了什么（what）。然而，当作者准备向外界报告他们的研究时，不能理所当然地认为该研究对读者来说是不言自明的，他必须有意识地思考并假设读者的知识水平，适当地补充相关知识。鉴于目前越来越多的多学科交叉的科学研究性质，作者在文章中做到这点尤其重要。希望获取研究成果的研究人员和利益相关者的人数范围是无法预计的，并且这个范围还在不断增长。

由此看来，作者所写的文字不是信息的资料库，也不是靠字句堆积出的水到渠成的产物，而是一件为读者创造的工艺品。**目的不仅仅是使读者能够理解，而且它的叙述应该强有力到使读者不可能理解不了。**

叙述性包装是什么?

为了交流研究成果，必须要让科学成为一种语言。叙述性包装是一种围绕信息沟通的脚手架，能够使读者更加明白易懂。决定题目、小标题、段落功能、图表的形式和顺序，这些都是构成叙述性包装的一部分，它们为需要表达的信息建立起一个有意义的框架。在写作中，这种叙述可以是通过作者评论和确认信息功能的方式来实现，也可以通过不太明显的方式，例如段落起始句或动词时态的选择来实现。这本书将讨论用叙述的形式包装信息时要用到的工具和指导思想。**叙述性包装将科学信息和数据变成让读者津津乐道和高效阅读的文本。**



规划：叙述方式脚手架的基础

确定文章的类型和读者群体：我要创作一件什么样的作品？研究论文、综述论文、项目申请书、快报、海报和会议摘要都属于不同类型，每一种类型需要不同的结构和表达内容的方式。在不同类型的作品之间剪切粘贴的方式将会产出一篇杂乱无章、毫无重点的文章。在开始规划你的论文之前，最好将相关类型的论文进行逆向工程以得到相应的写作模板。

谁可能会读这篇文章？我如何定位我的文章中信息的层次？我能假设所有的潜在读者都知道什么样的相关知识呢？我能以多快的速度将读者带到我想陈述的科学细节中呢？这篇文章的读者群会在日后变得更加跨学科吗？

小节划分：该领域同类型的论文有小节划分的标准吗？我的目标文章通常有多少个小节呢？每个小节的平均长度是多少？我的文章能有逻辑地精准匹配到这些小节中吗？如果不能，我有更好的方式划分小节吗？

选择图表（数据、表格、图片）的数量、顺序和位置：我有好的理由解释所包含的每个图表吗？我的图表最有逻辑的顺序是什么？每一个图表放在哪个位置最有助于读者？将每一个图表放到被相关文字围绕的地方，不至于打断文章叙述的流畅性。在叙述中，当读者意识到想看到图表的时候，使图表恰好自然地出现。

规划段落的功能和顺序：我所有的段落都有明确的功能吗？段落的功能对读者而言是清晰的吗？段落的顺序合乎逻辑吗？

规划句子在段落内的功能和顺序：我所有的句子都有明确的功能吗？句子的功能对读者而言清楚吗？或者说读者会质疑我写它的理由吗？每一个句子与段落的总体功能匹配吗？每一个句子和下一句在逻辑上都是连贯的吗？

写作：主动的叙述性包装

为读者评论信息和数据 我有（可能是错误地）假设我对信息和数据的理解对所有潜在读者都是明白易懂的吗？

为读者明确信息和数据的功能 我有（可能是错误地）假设我的信息和数据的功能对所有潜在的读者都是明白易懂的吗？

选择有助于传达信息功能的语法 我有考虑过每一个句子的动词时态的含义和影响吗？我选择的语法是否解决（或者产生）了潜在的歧义？

选择可以帮助读者的词汇 我有考虑过全球读者的需求吗？我有连贯性地使用词汇吗？

选择适当的起始段落和句子以帮助读者 每个段落的起始句有帮助读者看到这个段落的走向吗？我是否已经缩小了句子之间在含义上的差距，并以方便读者阅读的方式开始每个句子？

创建有助于读者的标题和小节标题 我的标题表现和预示了文章的内容了吗？还是说它仅仅是我在项目开始时没有细想的原始的“作品”标题呢？我的标题符合这类文章的通用长度吗？我的小节标题是否适用于整个小节？我的小节标题能帮助读者进入小节部分吗？这个小节的所有内容都“符合”标题吗？我的小节标题符合这类文章的通用长度吗？目标文章中的小节标题符合通用的语法吗？它们是否应该能概述相应小节的内容？

本书使用了什么策略

经典的写作方法是基于体裁的方法，该方法能处理不同类型文本的结构和内容。这种体裁教学法被广泛用于写作教学。然而，虽然这种方式有助于告诉作者他们需要做什么，但并不能很有效地告诉他们该怎么做。例如：

> 撰写讨论部分：
>
> Highlight the most significant results, but don't just repeat what you've written in the Results section. 我如何做到这一点呢？ Show how these results relate to the original question. 怎么做？ Discuss whether or not your results are consistent with what other investigators have reported. 告诉我怎么做！ Look at alternative ways to interpret your results. 如何做到这一点？ Discuss how your results fit into the big picture. 我究竟如何“讨论”这一点？

为了回答这类问题，本书的读者将看到如何开展逆向工程的方法。本书指导读者解构当前在各自领域成功的研究写作，为自己的写作创建模板。我曾在伦敦帝国理工学院和世界各地的其他大学使用过这种逆向工程法，分析了一系列 STEMM 学科中成千上万最新发表的研究文章，以构建和验证本书的基

本模板、语法和语言列表。伦敦帝国理工学院学术英语中心使用这种方法向早期研究人员教授 STEMM 写作，并培训更有经验的科研作者开发和应用这种方法。逆向工程法的一个关键优势在于它对研究写作日新月异的特性做出了很好的响应：一旦熟悉了这种写作策略，作者可以在下一轮科技写作和发表平台变更的同时调整和更新其模板。

本书的目的是为用于发表的科学研究写作提供一个快捷自助指南。逆向工程法可以被快速理解、独立使用，并适合所有 STEMM 研究学科和所有 STEMM 文本的需求，包括项目申请、会议摘要和行业报告。这一方法是描述性的而不是规定性的。它并不提供建议或告诉作者做什么和不做什么，其目的首先是帮助作者生成对他想要创作的写作类型的高度准确的描述。然而，仅凭准确的描述并不会产生文本——描述某件事与做成某件事并不相同。因此，除了描述如何做，这种方法的关键作用还在于它可以帮助作者发现如何操作以及如何应用该知识。

策略如下：

1. **建立基于句子/基于段落的能指明成功文本的功能的模型，**例如：*这句话指出了研究中的空白*（This sentence identifies a gap in the research.）或*这一段为本研究的贡献与文献建立关联*（This paragraph maps the contribution of this study onto literature.）。

2. **挖掘成功文本中表达功能的词汇，**例如*表达研究差距的单词和短语*。

3. **识别并掌握实现这些功能的相关语法、语言和写作技能，**例如动词时态的选择。

4. **阅读领域内的文章，不断增强、调整、更新和发展上述三点。**

上述过程中最后一步：**阅读领域内的文章，不断增强、调整、更新和发展上述三点**是必不可少的。科技写作不是一成不变的类型，它正不断地演变，成功的作家需要有意愿且有能力随着类型的发展而升级。成功的关键是参考一系列目标研究文章（或综述论文、会议摘要等），不断更新上述内容和相应的模板。

每章中介绍的模板、语法/语言和词汇是对 STEMM 范围内的2500多篇最近发表的研究文章进行分析的结果，以生成可靠且具有高表面效度的工具包。

关键信息：

- **科技写作的内容应该是面向所有读者的。**
- **作者需要向读者展示数据或信息的功能，而不仅仅是数据或者信息本身。**
- **写作的目的不仅仅是使读者能够理解，而且是使读者不可能理解不了。**
- **研究已成功发表的论文的结构和语言以获得循证指导。**

本书面向哪些读者群?

本书旨在使科学研究者（包括英语母语人士和非英语母语人士）撰写更有效的论文以供发表。书中描述的逆向工程策略也可以用于其他与研究有关的 STEMM 写作，例如报告、会议摘要和综述文章。此外，由于研究论文是学位论文的缩影，因此它也可以用来指导撰写博士学位论文。事实上，许多机构都允许博士候选人以一系列相关的已发表论文的形式组织和提交学位论文，因为这同时解决了撰写学位论文和相关研究论文的问题。

我为什么需要这本书?

科学在发表之前是不存在的[2]。好的写作会增加发表的机会，并且发表是进行研究的一个主要目标；当你通过发表文章来分享你的工作成果时，你将成为全球学术界的一员。此外，高质量的已发表作品能让你获得基金的支持，并引导你参与高水平的项目。撰写和发表研究论文也是事业发展的最佳途径。如果你将毕业论文或项目变成有用的论文，那么你的简历看起来将更专业，并且在国际上更具竞争力。

然而，优秀的科学家却不一定总是优秀的作者。有些人发现自己很难以最佳的方式组织信息，或简洁有效地展示他们的研究。其他人则觉得写作很麻烦，并且可能想到要写作就会感到气馁。这会减慢写作的速度——有时会完全停止。在当前全球性的背景下，研究工作需要迅速发表。本书旨在通过提供你

所需的信息、词汇和技能来加快研究成果“从实验室到期刊”的过程，以便你可以快速、自信和独立地进行写作。

作为一名科学研究员，你能够无障碍地阅读和理解你所在领域的高度复杂的材料。然而，你发现自己却很难撰写一篇与你阅读的文章水平相当的论文，或者你可能会感觉到你的文章不能有效地传达你的研究内容和研究的影响力。在本书中，你将学习如何利用现有的阅读能力和所阅读的材料来发展相关的写作技巧。你不需要学习更多的知识，只需要有效运用自己已经掌握的知识，而不是想到哪写到哪。科技写作要比看起来的更容易。

为什么不简单地使用校对服务，或者（非英语母语的研究人员）使用翻译软件?

向校对员或翻译寻求帮助可发现一些语言错误，但可能无法解决写作问题。他们可能不会注意到某些错误，例如一个语法上完美的句子仍可能是“错误的”，因为它未能传达你的本意。例如，错误的动词时态可能会使你通过研究发现的信息（X occurred）看起来像已知的和被接受的事实（X occurs）。此外，校对者或翻译者可能无法检查你的写作是否符合你所在领域的传统模式。例如，你可能忘记解释你所选择的方法或者结论与原始问题的关系。校对或翻译人员不可能留意到这点，但这可能导致同行审稿人或期刊编辑认为你的论文不够专业而拒绝录用。另一个例子是，如果你的结论被翻译成比你预期的更

有力或更具肯定性的语言，那么你的论文可能会在评议阶段失败，因为审稿人无法在你的结果中找到足够的理由，以说明你能得出如此的结论。

使用人工翻译或翻译软件还会带来另外的困难。在词汇选择上，甚至在将信息组织成段落的方式上，英语学术写作遵循着某些文化规范，这些规范都是明确的。如果作者与读者的文化写作规范有很大差异，那么翻译后的文本就会显得莫名其妙，并且论文的某些关键方面如作者想指明研究价值的部分，就可能被低估以至无法被认可。另一个很快会出现的问题是，一旦把机密材料交给互联网的翻译软件或编辑工具，作者就很难控制其所有权。

这本书能帮助我成为论文的“第一作者”吗?

第一作者在文献中比共同作者更引人注目，并且与研究更加紧密地联系在一起（例如，Wu *et al.*, 2019）。在不同的研究学科中，针对多位作者的顺序，规则和惯例各不相同。但是，如果你靠同事完成了大量写作，这可能会影响你作为第一作者的地位。因此，本书的主要目的是提升你独立写作的能力。

这本书的重点是词汇和语法吗?

科学是没有国界的，这影响了科技写作的语言，并促使英语母语的作家和非英语母语的作家用对于世界各地读者而言都清晰易懂的语言来交流他们的研究。无论你的母语是什么，大

多数阅读你论文的人都不是英语母语者。因此，科学研究写作需要使用相当有限的语言来表达和交流具体的含义。对于非英语母语人士来说，这是个好消息！这意味着他们不必掌握全部英语；要成功撰写清晰的论文，只需要掌握一小部分英语词汇即可。在研究论文中用于交流的语言通常是由一组有限的常用单词和短语组成，科技写作中使用的非技术词汇也只包括一组有限的单词和短语。这些词语被当作是一种代码，可以与世界各地的研究人员进行准确的交流。

语法问题是更加复杂的。如上面所述，正确的语法不能确保沟通的准确或有效；同样，尽管存在着语法错误的文本也可能写得很好，可以成功且准确地进行交流。本书并不着重于语法规则，而是指导所有作者使用语法——例如动词时态——以清晰、明确地传达预期的意思。

我为 STEMM 学生教授学术英语课程已经有30多年了。在过去的25年中，我一直在伦敦帝国理工学院的学术英语中心为博士生和博士后教授科技论文写作，同时我为个别的博士生、博士后和学术人员提供咨询，其中包括英语母语和非英语母语人士，他们都正在撰写学位论文、研究论文或者教科书。我还为世界各地的研究小组举办研讨会。本书基于我所学到的两个最有用的内容：

1. **科技写作不需要时尚或优雅；其主要目的是清晰、准确地沟通。**使用雅致的英语实际上可能会使你的研究文章难

懂，甚至更不准确。例如，使用避免重复的同义词库会适得其反，未经预先告知，框架（framework）不应突然成为方法（methodology）、途径（approach）或模型（model）。重复相同的单词会让读者产生共鸣，使他们能够识别并跟踪论文中的概念，并确保概念本身的稳定和一致性。

2. **良好的组织和优秀的写作可以弥补语言错误，但毫无错误的语言不能弥补不良的组织或糟糕的写作。**如果信息混乱或呈现得不连贯，那么使用语法完美的英语撰写的文字也将是无效甚至是难以理解的。

这本书如何展开？

本书以研究文章作为 STEMM 文本的典型示例。报告、专题论文和学位论文的构建方式是类似的，但研究论文更多，更容易进行分析。

为了简单起见，本书中的单元根据 IMRaD（引言、方法、结果和讨论）进行排序。尽管 IMRaD 不再是研究写作的唯一可用模式，但大多数研究文章的确包含引言、已完成/已使用内容的描述、已完成/已使用内容的结果以及对该结果的讨论这些部分。然而，IMRaD 不是作者通常写研究论文的顺序，也不总是论文在刊物中出现的顺序，也可能不是阅读的顺序。作者通常首先写方法部分，然后是结果，接着是讨论/结论，之后是引言，最后才是摘要和标题。此外，一些刊物将方法或实验放在论文的末尾，或将其作为补充材料。读者也很

少按 IMRaD 的顺序进行阅读，他们可能以标题/摘要（Title/Abstract）开头，然后根据其兴趣直接跳到结果（Results）或方法（Methods）部分。尽管如此，为了方便参考，本书中研究论文的各个部分还是以 IMRaD 格式显示，并展示其通常在刊物中出现的顺序。

本书分为八个单元。第一单元是引言，第二单元是方法，第三单元是结果，第四单元是讨论，第五单元将前面单元中使用的逆向工程方法扩展到结论部分，第六单元是摘要，第七单元是标题，第八单元包含一个清单和写作技巧。

前两个单元的结构相似。例如，第一单元首先建立一个关于引言的通用模板，该模板来源于最近发表的论文的引言，并演示该模板在实际应用中的运作方式。接下来是一系列选自不同领域的最新发表的2500多篇研究文章的有用词汇，以及一些句子的模式示例。这些示例显示了如何在引言中使用单词和短语。该单元的末尾设有一个语言和写作技巧部分，旨在回答常见的问题。每个单元末尾的语言和写作技巧部分没有太多的语法练习；本书旨在作为一本**快捷且易于访问的指南**，而不是一本语言教科书。在讨论语法问题这方面，其目的不是确认“正确”语法的规则，而是避免歧义。

在第一单元的这一阶段，你将拥有一个强大的引言通用模板，关于模板每个部分有用的单词、短语和句子模式的列表，以及语言/写作技巧指南，并对引言中有关语法和写作技巧的常见问题做出回答。如果你能在本单元中完成任务，则可以将

模板、词汇和语言/写作技巧连接到一起编写一个引言。在第一单元的最后，你将使用所学的知识编写一个简单的引言，并将其与答案中的示例进行比较。第二单元重复此模式。

从第三单元到第六单元的内容都鼓励你对自己的目标研究文章采用逆向工程方法去创建模板，然后根据每个单元中提供的通用模板对它们进行检查和调整。本书中的所有任务旨在帮助你开发逆向工程，使其成为你阅读策略的自动辅助方式。如果阅读了本书却没有完成练习和任务，你将对所从事的写作有一定的理论认识，但是你可能会发现很难将其付诸实践。知道该怎么做与知道如何去做是不同的。目前，你可能非常关注研究文章的信息内容，而没有过多关注内容的发布方式。但是，一旦你察觉到用于传递内容的结构和语言，就会创建一个“学习闭环”，它能使你通过科研阅读主动和连续地为研究写作提供帮助。

我还需要什么其他材料或书籍？

作为一名科学研究人员，你需要以与你所在领域当前成功发表的作品尽可能相似的方式进行写作。这本书向你展示了如何利用你阅读的期刊论文来发展适当的写作技巧，因此在开始之前，你需要收集一组目标研究文章。这组论文应至少包含你通常阅读的期刊中最近发表的四篇研究论文。你将用你所学到的知识把这些目标研究文章的写作方式应用到你自己的写作，并且你应该经常更新这组论文。

你要收集的目标研究文章应该：

- 由一个在说英语的研究机构的研究者或研究团队撰写，最好是英语母语人士。
- 所涉及的主题要与你正从事的研究类型尽可能相近。
- 精练（包括图式和表格，不超过15张 A4纸）。
- 摘自你渴望发表文章的期刊。
- 不超过5年以前（鉴于科学研究写作的发展和变化速度）。
- 是你认为写得优秀的研究文章*。
- 有明确定义的引言（Introduction）、方法（Methods）、结果（Results）、讨论（Discussion）和结论（Conclusion）部分，或其他与你正在从事的作品相符的可识别部分。如果这些文章有小节标题，将有利于你快速定位。要注意这些小节标题在不同领域和不同期刊上也会有所不同；比如 Methods（方法）也可叫作“Procedure”（步骤）、“Model”（模型）、“Materials and Methods”（材料和方法）、“Methodology”（方法论）、“Experimental”（实验部分）、“Data Collection”（数据收集）或某些其他变体。

* 对于这一点，我要向莫妮克·多朗热（Monique Dorange）教授表示感谢。

第一单元
怎么写引言

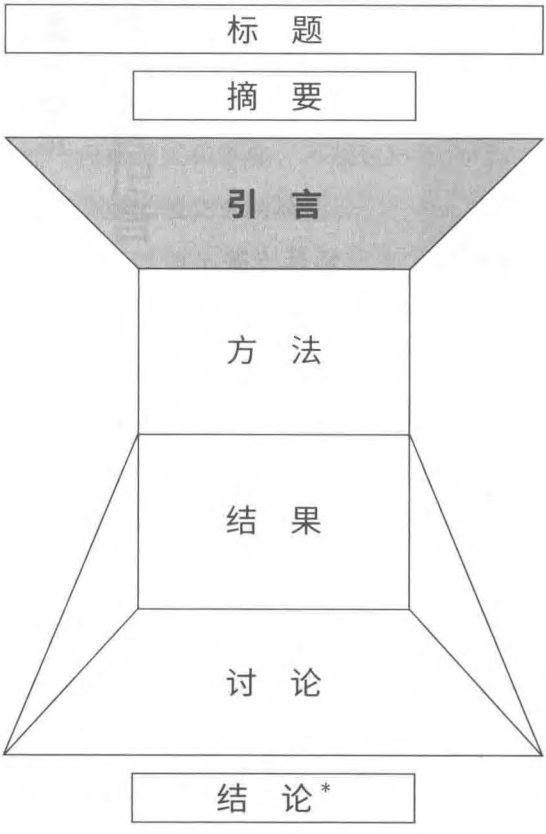


图 1.1 一篇研究论文的结构形状

* 有些期刊在这一部分使用结论一词的单数形式，有些使用复数形式，但这似乎并不能反映所得结论的数量。

> 不幸的是，在一个许多人为了自己的职业而必须发表论文的环境中……作者只有很少的精力是花在理解论文的结构上的。这显然是一个需要学习的重要内容。潜在作者需要详细的、循证的指南，并且该指南要能够通过研究成功发表在主流期刊上的论文的结构来获得。[3]

1.1　引言的结构和内容

如你所见，图1.1是对称的。这是因为引言中的许多事情也经常会出现在讨论部分（通常是以相反的顺序）。例如：

- 在引言的开头，你写一个开篇句子使读者能够“进入”或开始阅读论文/学位论文；在讨论结束时，你会通过找个合适的结束方式“走出来”。
- 在引言中，你通常会介绍有关该主题的过去或当前的研究/知识；在讨论中，你将展示你的工作如何为该研究做出贡献或增进知识。
- 在引言中的某个地方，你陈述了你的工作重点关注的特定问题，或者指出了现有研究中需要填补的空白；在讨论区，你将阐述你在多大程度上解决了该问题。
- 在引言的最后，你需要使读者能够进入文章的中间部分（通常是“方法”和/或“结果”）；在中心部分的末尾，你需要使读者能够进入讨论部分。

需要注意的是，图1.1的形状在中间部分变窄，而在讨论部分又变宽了。这表示在引言和讨论中对信息进行组织的方式：在引言部分，你从相当宽泛的角度开始，逐渐将重点放在自己的研究，而在讨论部分则相反。

1.2 创建模板

引言的结构在大多数研究领域中都相似，而内容与结构直接相关。本单元向你展示如何构建和使用引言模板，该模板回答以下三个问题：

- 作者通常如何开始引言？
- 引言中通常包含哪些内容？顺序如何？
- 作者通常如何结束引言？

1.2.1 练习1：如何通过逆向工程构建简单的模板

利用本书引言中“本书使用了什么策略”讨论的逆向工程方法，查看下面的第一个引言，并简短描述作者在每个句子中做什么。请注意，不是要描述或总结句子在说什么（句子的内容）；而是要弄清楚句子在做什么（句子的功能）。识别句子功能的一种方法是查看主要动词的时态。这一句的动词与前一句的时态相同吗？如果不是，作者为什么改变了时态？

请记住，你的模板仅在可以转移到其他引言时才有用，因此不要加入如“聚合物”之类的内容词，否则你将无法使用该模板为自己的研究文章生成引言。

保持描述简单，句7已为你完成了你要达到的详细程度的指南。一旦完成了第一个引言的描述，请将该模板应用于1.3.1节中的引言，然后再将其应用于你的目标文章中的引言来进行测试和调整。

A light-weight chitin-PLA composite for load-bearing biomedical applications

Introduction

***1** Biomass-derived polylactide (PLA) has received much attention in recent years due to its biodegradable[1] and biocompatible[2] properties. **2** Biomass-derived PLA is produced from renewable feedstocks such as corn and sugar cane. **3** It has outstanding molding properties for use in biomedical applications such as scaffolds for tissue engineering[3,4]. **4** However, its weakness under impact has significantly limited its use.*

***5** Biomedical applications require materials with the lowest possible toxicity so although metals such as aluminium have been used to strengthen PLA for industrial applications[5,6],these are not appropriate for biomedical use[7]. **6** One way to strengthen polymers for biomedical applications*

is to incorporate a layer of keratin, a material that occurs naturally in the mammalian anatomy, and there has been extensive research regarding keratin-PLA composites[8,9,10]. 7 For example, Penney et al. (2012) showed that incorporation of keratin fillers in the PLA matrix could be achieved using blending techniques[8]. 8 However, although the effect of keratin on the mechanical properties of copolymer systems was demonstrated over two years ago[9], keratin adds considerably to the weight of the composite, and little attention has been paid to the selection of an alternative biocompatible material.

9 The present paper presents a set of criteria for selecting such a component. 10 On the basis of these criteria it then describes the preparation of a biomass-derived polymer blend using chitin, a long-chain polymer found in many places throughout the natural world. 11 This combination formed a novel lightweight copolymer in which the incorporation of PI significantly increased resistance to impact while retaining the biocompatible features of keratin.

在第 1 句，作者	
在第 2 句，作者	
在第 3 句，作者	
在第 4 句，作者	
在第 5 句，作者	
在第 6 句，作者	

在第 7 句，作者	提到本研究领域的一个关键研究
在第 8 句，作者	
在第 9 句，作者	
在第 10 句，作者	
在第 11 句，作者	

1.2.2　答案

句1："Biomass-derived polylactide (PLA) has received much attention in recent years due to its biodegradable[1] and biocompatible[2] properties." **作者确立了该研究主题的重要性。**

如果你把句1总结为"**引出主题**"，这对你撰写自己的研究论文没有帮助，因为它没有告诉你如何"引出"这项工作。你可能会通过描述你要解决的特定问题或者计划使用的方法来开始以前那种引言，但是当你查看你的领域中已发表的论文时，你会发现大多数研究论文不是用这样的方式开始的，因此这可能不是最佳的开始方式。许多研究论文首先要论证这一主题如何或为什么有用或重要。因此，在你的目标文章中，你通常会在第一句话中发现一些表达方式，如much study in recent years（近年来的许多研究），或者a major role or widely-used（起重要作用或被广泛使用）。

我在第一句话应该使用什么动词时态?

对于引出近年来的工作的短语,如 much study in recent years(最近几年)或in the past five years(在过去五年),后面通常用现在完成时态(如“ Much study in recent years has focused on.../The status and function of the part has evolved rapidly over the past three years.”)。如果你关注当前的情况,通常使用现在时态(如“There are substantial benefits to be gained from...”)。但是,值得注意的是,现在或当前的情况将不会在五年后成为最近或当前的情况。因此,如果你希望你的研究文章五年后仍然具有相关性,可以考虑使用可识别的日期,例如 since 2018(自2018年以来)。

如果我没有信心说我的研究是重要的该怎么办?

如果大多数研究论文的作者通过描述研究的重要性来开篇,而你不是这样开篇,会让编辑和读者看起来感觉你的研究主题并不重要。检查你的目标期刊——这不是信心问题,它是关于以一种公认的传统方式进行写作的问题。如果你所在领域的大多数目标文章都是以这种方式开始的,那么你也应该这样做。所以,不要害羞,大胆地说你的研究是重要的或者有用的。

所有的研究论文都是这样开始的吗?

不是的,目标读者群始终是决定因素。如果一些研究主题非常具体,所有想阅读你的研究论文的人都是一个小型科研

团体的一部分，那么就没有必要对他们声明该研究领域的重要性，这样可能是不合适的。在这种情况下，通常先从对有关该领域研究的特定重点的陈述、关键术语的定义或理解该主题所必需的一些事实信息开始。句2就是这种表达的一个例子：

句2："Biomass-derived PLA is produced from renewable feedstocks such as corn and sugar cane." **作者为读者提供了背景事实信息。**

句2使用一般现在时，这个动词时态用来表示已经被接受/确立的事实。学术论文通常在第一句或第一段，以既定事实开篇。这就确保了所有的读者都有适当的背景信息来处理论文的其余部分。

那么我应该以什么样的事实开头呢?

以什么样的事实开头取决于你的主题的宽度——同时这也取决于你的读者群。如果你的研究范围很窄，那么你的很多读者就会有相当高的背景知识，你可以从比较具体的信息开始。如果你的论文可能会吸引更广泛或更跨学科的读者，那么就要以较宽泛的背景信息开始，甚至考虑包括对你来说很明显的背景事实。你可以查看**目标期刊上最近发表的类似论文**，找到有关第一个事实的良好示例。

请注意，如果你将论文重新提交给其他期刊，那么该期刊的读者群可能会很窄或者更宽，因此你不能简单地重新发送原始版本。特别是你可能需要调整第一句话，以确保它能满足

不同读者的需要。黄金法则是：**检查你要提交的期刊的最新版本**，并以和其他在该期刊上成功投递过类似主题的作者相同的方式开始你的引言。

如果有多个背景事实，那么我该从哪个开始呢？

图1.1中引言的形状开始宽而逐渐变窄，这意味着你要从一个相当宽泛的，许多读者已经知道的信息开始。这是一个类似于“集合地点”的事实，也就是你的所有读者都可以开始的起点，从这里开始进入更加具体的信息。总是要在讨论细节之前给读者展示概貌：就像在你给读者看砖头之前先把整面墙展示给他们。

我应该提供多少背景信息？

作者很难预测读者对论文的认知有多少。你的读者不是你的研究组的成员，因此你和你的同事非常熟悉的背景信息不一定是研究论文的潜在读者都知道的。鉴于现在大多数研究阅读都是通过互联网进行的，因此读者可能来自其他学科甚至非STEMM学科。通常，提供的背景信息多一些总比少一些的好。

此外，如果你在引言的开始部分从非常宽泛的信息突然跃升为非常具体的信息，这可能会对读者造成困难，特别是对于跨学科的读者。因此，弥合开篇句子之间的信息差距至关重要（参见第1.5.2节），以便所有读者都能顺畅浏览引言中的背景信息。

▶ 我还是不清楚该从何处开始。

如果你仍不知道应该怎样开始首句，那就看一下你的关键词。如果你的目标期刊中类似的近期论文都以其中一个关键词的定义或事实开头，则应考虑以关键词的定义或事实开始；如果他们首先确定主题的重要性，那么就以此开始。记住关键的信息：研究已成功发表的论文的结构和语言以获得循证指导。

▶ 我不能以我希望解决的问题开始吗？

大部分作者都不会这么做，因为很多时候只有当读者有足够的背景知识去理解问题的时候才能够把具体问题说清楚。另外，很难用一句话概括你要解决的问题。对你来说，在说清楚要解决的具体问题之前，你已经了解了很多具体的信息。但是对于读者来说，如果你没有给读者提供足够多背景知识，他们很难处理这些信息。

句3："It has outstanding molding properties for use in biomedical applications such as scaffolds for tissue engineering [3,4]."
作者的做法跟在第1句和第2句是一样的，只不过方式更具体，通过引用参考文献来支持背景事实和重要性声明。

▶ 句子3中有引用参考文献，是否意味着这是该领域研究综述的一部分？

不是的，引言的后面通常会有以前和现在的研究和知识的简短综述，并且更可能侧重于具体研究的方法或结果。这里只

是提供背景事实的参考文献。

▶ 如何确定是否需要提供背景事实的参考文献资料?

包含引用的原因之一是剽窃(未能尊重别人为作品付出的劳动)是不道德的;第二个原因是引用参考文献可以使读者有机会找到并阅读所引用的研究;第三个原因是,如果未能提供引用通常表明你对本领域的研究还不够熟悉。

是否包括文献的引用还取决于该工作在何时发表,以及领域研究团体将该研究工作的结果作为普遍接受事实的程度。有许多事实是研究人员已经广泛接受的(也许是多年以前就接受了)并且作为他们研究工作的起点。例如,句2中的信息"Biomass-derived PLA is produced from renewable feedstocks such as corn and sugar cane."来源于已有研究,可能最初是用现在完成时来写:"Lawrence *et al.* have found that biomass-derived PLA can be obtained from renewable feedstocks such as corn and sugar cane.",或者是使用过去时:"Lawrence *et al.* (2000) found that biomass-derived PLA can be obtained from renewable feedstocks such as corn and sugar cane."。但是,一旦该研究发现被充分确立,那么陈述就可以通过现在时来叙述,并附带引文:"Biomass-derived PLA is produced from renewable feedstocks such as corn and sugar cane (Lawrence *et al.*, 2000)."最终,当该信息变成了研究领域公认事实背景的一部分,引文就可以不需要了:"Biomass-derived PLA is produced from renewable

feedstocks such as corn and sugar cane."

正确执行此操作的唯一方法是**检查你的研究领域中当前的使用情况**。五年前的期刊文章所使用的动词时态到现在可能不再适用了，因此如果仍然使用相同时态就意味着你不了解该领域的最新动态。搜索引擎（例如Google Scholar）是一个很好的资源。如果你所在领域的大多数研究人员当前在陈述一个特定事实时使用现在时，则在你陈述该事实的时候也应该遵循他们的观点。如果大多数仍然包括参考文献资料，请遵循其指导。令人兴奋的是，一旦在你的领域确立了你作为已发表论文研究人员的地位，你就会获得自己做出此类决定的权利。

句4："However, its weakness under impact has significantly limited its use." **作者描述研究领域的一般性问题或者当前的研究焦点**（weakness under impact）。

请注意作者仍未开始描述本研究论文要解决的具体问题，而是还在叙述这一研究领域当前的焦点问题，例如，关于 weakness under impact 的问题。这些问题恰恰是本领域的很多研究者所关注的，并且会指引到本论文所要解决的具体问题。

句5："Biomedical applications require materials with the lowest possible toxicity so although metals such as aluminium have been used to strengthen PLA for industrial application[5,6], these are not appropriate for biomedical use[7]." **作者扩展了研究领域的问题范围/当前的焦点**（toxicity）。

句6："One way to strengthen polymers for biomedical

applications is to incorporate a layer of keratin, a material that occurs naturally in the mammalian anatomy, and there has been extensive research regarding keratin-PLA composite[8,9,10]." **作者将上面一般的问题和已发表的研究联系起来，并引用了其他研究。**

▶ 在句5中，一些文献引用出现在句子的中间。什么时候需要把文献引用放在句子的中间?

当需要避免混淆的时候，例如如果引文仅涉及句子的一部分，你的引文位置应清楚地说明哪一部分的信息来自哪一项研究。这不仅是一个职业礼貌问题，而且暗示一个句子中的所有信息都来自该句子结尾处引用的所有研究是一种误导。

句7："For example, Penney *et al.* (2021) showed that incorporation of keratin fillers in the PLA matrix could be achieved using blending techniques[8]." **作者提到该领域的关键研究。**

▶ 在引言中我应该提到多少项研究?

如往常一样，请参考你的目标研究文章。值得注意的是，如果你和许多作者合作撰写论文，或者论文本身需要花费很长时间去写，引言中的文献和知识综述可能会变得冗长而失去重点。你在引言中提到的每项研究都应该与自己的研究工作背景相关且必不可少，它们应引向你的研究动机。

如何确定引言中要提及的研究?

相关性是至关重要的。研究论文中的文献和贡献综述并不是要你列举或者总结你所阅读过的内容。这是**一个精心叙述选定的相关研究以及对研究有帮助的资料的过程**，其目的明确：是向读者证明论文报告的研究是合理的，并且这种方法是有效的。如果包括无关的研究，你将打断叙述的思路。决定要在引言中包含哪些研究工作直接关系到能否为你在论文中的研究提供动机的问题或疑问，因此，请选择能够说明问题的发展和演变的研究工作。

文献和贡献综述的另一功能是能够指明你的工作在该研究领域的“地图”上的位置。当你阅读研究论文时，通常会认识到引言中提到的一些研究，它们可以帮助你了解该工作在研究领域中的位置。同样，你在引言中提及的名称和参考文献将帮助你的读者了解你的工作在研究领域“地图”上的位置。

我如何组织和连接这些研究?

整理和组织这些研究并将它们与理性的叙述联系起来是必不可少的。你不能在没有连接叙述的情况下就将以前的和现在的研究的描述毫无顺序地“倒入”页面，否则它看起来就像是一张购物单。细心计划和设计过的文献和贡献综述有助于读者理解研究的发展状况，更重要的是有助于读者理解它们是如何以及为什么促进你的研究的。

为了保持引言中的综述不离题，需要根据一个有组织的模

式来计划和组织研究的贡献。这里给出三种可能的模式：

- 从一般到具体：作者经常从该领域的宽泛研究开始，可能是一篇综述论文，然后逐渐转向更接近本研究中所解决的问题或当前研究的空白。这是最常见的模式。
- 按不同的方式/理论/模型：在一些研究中，根据不同的途径或方法来对项目或研究进行分类。把同类的项目归到一起能帮助你避免“网球”式来回运动的效果，每个句子可以用 However 或 On the other hand 来开始。
- 按时间顺序：如果你研究的领域同污染或水力压裂法方面的政治决策和法律有关[①]，这种顺序也许是合适的。

为了将研究和贡献联系起来，可以考虑如下的一些单词和短语：

[ref] also observed that…
[ref] resolved this by…
[ref] were the first to…
A pioneering study by [ref] demonstrated…
A similar approach was used by [ref], who…
An alternative approach was proposed by [ref], who…

① 原文在此处的表述简单而含糊，其仅使用一个单词“fracking”指代开采页岩油常用的“水力压裂法”。相关开采活动因会破坏生态环境而备受争议。——编者注

According to [ref], it is highly likely that…

Importantly, [ref] noted that…

In order to resolve this, [ref] analysed/developed...

In that study, although…

In their recent work, [ref] has suggested that…

It was later shown by [ref] that…

One [potential] implication of [ref]'s results is that…

Other studies have focused on…, for example, [ref] and [ref] attempted to…

Recently, it has been shown/suggested that…

Soon after, [ref] proposed…

Subsequently,…

Taken together, these studies suggest that…

The most systematic review is that in/of [ref]…,

These findings challenge the work of [ref], who…

This approach was further developed by [ref], who…

This methodology was adapted by…

To address this issue, [ref]…

To overcome these problems, a different approach was used by/in [ref].

Within five years, [ref] developed a…

像however/therefore这样的信号性语言也可以有效而简洁地将叙述从一项文献和贡献综述带入另一项。有关更多信息，**请参见第1.5.2节：句子和信息的连接。**

句8："However, although the effort of keratin on the mechanical

properties of copolymer systems was demonstrated over two years ago[9], keratin adds considerably to the weight of the composite, and little attention has been paid to the selection of an alternative biocompatible material." **作者指明研究的问题或空白。**

从这里开始你转向自己研究的论文或工作，以及将要解决的具体问题。通常，这可以通过确定现有知识的空白或通过指出已有研究中存在的问题来进行。作者往往会使用诸如 However 或 Although 这样的标志性连接词来引出这些问题或者研究空白。专业写作中很少会用提问题的形式来引出空白或者问题，通常可以把问题描述成你打算验证的猜想、建议或者假设。

不要羞于指出以前研究中的问题。首先，识别出以前研究中的问题是很有必要的，它可以解释你做现在这项研究的原因。其次，这里的语言往往是恭敬且客观的，因此不会被认为是冒犯别人的。

在此阶段，读者可能需要更多背景信息。例如，你需要更具体地描述所要研究的材料性质的详细信息，或者计划改进的设备的具体部分。研究写作需要数量惊人的背景知识，而提供更多的背景信息总比背景信息太单薄要好一些。

在这个阶段你就开始转到目前的论文和研究。你可以描述它，说明此论文和研究的目的或重点是什么，呈现它的结构或者以上这些信息的综合。

句9："The present paper presents a set of criteria for selecting such a component." **作者指明当前的论文或研究工作。**

我应该在这里使用被动语态还是主动语态?

这取决于诸多因素，如产生歧义的潜在可能性、目标期刊的风格以及你是更专注于作为作者的自己还是研究本身。相关的详细信息和指导，请参见**第1.5.3节：被动和主动语态的选择**。

我应该使用什么动词时态?

作者通常会用一般现在时来描述工作本身（如 This paper is organised as follows/This study focuses on...），而用一般过去时来说明研究目的（The aim of this project was.../Our aim was...），因为在“真正的时间”里，目的都是在工作开始之前确定的。也可以用一般现在时来呈现研究目的（The aim of this work is...），特别是在仅仅实现了部分目标的情况下。

句10：“On the basis of these criteria it then describes the preparation of a biomass-derived polymer blend using chitin, a long-chain polymer found in many places throughout the natural world.”**作者概述了论文中的方法**。

尽管作者经常在引言中提到研究方法或结果，但是如果你提供的详细信息过多，引言可能会变得很冗长，并且失去重点。与往常一样，查看当前的目标文章以获取指导。

句11：“This combination formed a novel lightweight copolymer in which the incorporation of PI significantly increased resistance to impact while retaining the biocompatible features of keratin.”**作者使用明确标识研究成果的语言来宣布主要结果，**

这里用了“高兴”的字眼：novel（新颖）、lightweight（轻巧）、significantly increased（显著增加）。

1.2.3 引言模板

这里列出我们在1.2.2节中通过逆向工程在引言中收集的句子描述：

句1，作者	论证了研究课题的重要性
句2，作者	为读者提供了背景事实信息
句3，作者	跟在第1句和第2句的做法是一样的，只不过方式更具体，通过引用参考文献来支持背景事实和重要性声明
句4，作者	描述研究领域的一般性问题或者当前的研究焦点
句5，作者	扩展了研究领域的问题范围/当前的焦点
句6，作者	将上面一般的问题和已发表的研究联系起来，并引用了其他研究
句7，作者	提到该领域的关键研究
句8，作者	指明研究的问题或空白
句9，作者	指明当前的论文或研究工作
句10，作者	概述了论文中的方法
句11，作者	使用明确标识研究成果的语言（“高兴”的字眼）来宣布主要结果

把这些内容进一步精简，就可以得到引言模板的四个基本组成部分。撰写引言时不必使用所有内容，该模板可以被视为灵活的菜单。

通用引言模板	
1	**论证研究主题/领域的重要性** **给出背景事实和信息** **给出研究领域的一般性问题/当前研究焦点**
2	**呈现已有研究和/或当前的研究和贡献：**研究“地图”
3	**确定已有研究工作的空白** **描述你要解决的问题** **呈现你的动机和/或假设** **确定研究机会**
4	**描述当前的论文，**有时会提及目标/结果/方法/结论，并经常包括“高兴”的字眼

当你看过很多篇引言之后，你会注意到：

- 几乎所有研究文章的引言都始于模板的第一组。
- 几乎所有研究文章的引言都以模板的第四组结束。
- 在一些引言中，模板的第二组是当前知识的概要。通常使用一般现在时，但可能仍需要引用参考文献。
- 在一些引言中，模板的第三组中的研究空白或问题没有明显地说出来。这是因为研究的动机可能只是扩展已有的研究范围，或者是研究空白/问题隐含在对当前

研究及其目标的描述中。

- 以一般现在时态表述的补充背景事实信息可以出现在作者认为有必要的任何地方。

1.3 验证和调整模板

1.3.1 模型的示范

为了演示该模板，在下面的引言中以**粗体**显示模板组成部分的线索。仔细阅读以下引言，并注意词汇和动词时态是如何与模板联系起来的。

Rapid fragmentation of neuronal networks at the onset of propofol-induced unconsciousness[a]

Introduction

General anesthesia **is** a drug-induced reversible coma **commonly** initiated by administering a large dose of a fast-acting drug to induce unconsciousness within seconds **(1)**. This state **can be** maintained as long as needed to execute surgical and many nonsurgical procedures. One of the most **widely used** anesthetics is propofol, an i.v. drug that **enhances** GABAergic inhibitory input to neurons **(2–4)**, with effects in cortex, thalamus, brainstem, and spinal cord **(5–7)**. **Despite** the

understanding of propofol's molecular actions, it is **not clear how** these effects at molecular targets affect single neurons and larger-scale neural circuits to produce unconsciousness.

The effects on macroscopic dynamics **are** noticeable in the EEG, which **contains** several stereotyped patterns during maintenance of propofol general anesthesia. These patterns **include** increased delta (0.5–4 Hz) power **(8, 9)**; increased gamma (25–40 Hz) power **(9)**; an alpha (~10 Hz) rhythm **(10–12)** that **is** coherent across frontal cortex; and burst suppression, an alternation between bursts of high-voltage activity and periods of flat EEG lasting for several seconds **(13, 14)**. In addition, slow oscillations (<1 Hz) **have been well characterized** in deeply anesthetized animals and **are associated** with an alternation of the neuronal membrane potential between UP (depolarized) and DOWN (hyperpolarized) states **(8, 15)**.

Although these patterns are observed consistently, it is **unclear** how they are functionally related to unconsciousness under general anesthesia. **Most studies have focused on** a deep steady state of general anesthesia **and have not used** a systematic behavioral measure to track the transition into unconsciousness. This steady-state approach **cannot** distinguish between patterns that are characteristic of a deeply anesthetized brain and those that arise at the onset of unconsciousness. Unconsciousness **can occur** in tens of seconds **(4)**, but many neurophysiological features **continue** to fluctuate for minutes after induction and **are** highly

variable between different levels of general anesthesia **(1, 16)**. Therefore, identifying the specific dynamics associated with loss of consciousness (LOC) **requires** an examination of the transition into unconsciousness, linking neurophysiology with behavioral measures.

In addition, the dynamic interactions between cortical areas that underlie these EEG oscillations are **not well understood**, because **few studies** have simultaneously recorded ensembles of single neurons and oscillatory dynamics from sites distributed across the brain. Consequently, how propofol acts on neural circuits to produce unconsciousness **remains unclear. A leading hypothesis suggests that** anesthetics disrupt cortical integration **(17, 18)**. Identifying the mechanism by which this disruption might occur **requires a better understanding of** how the spatial and temporal organization of neural dynamics evolves during induction of unconsciousness.

To address this question, we investigated both neuronal and circuit-level dynamics in the human brain during induction of unconsciousness with propofol. **We obtained** simultaneous recordings of single units, local field potentials (LFPs), and intracranial electrocorticograms (ECoG) over up to 8 cm of cortex, enabling us to examine neural dynamics at multiple spatial scales with millisecond-scale temporal resolution. **We used** a behavioral task to identify within seconds the time at which patients became unresponsive to auditory stimuli, which we defined as LOC.

Our results reveal a set of neurophysiological features

that accompany LOC that, together with previously reported effects **(8, 9, 15)**, enable a multiscale account of this profound shift in brain state. **We find that** LOC **is marked** by the abrupt onset of slow oscillations (0.1–1 Hz) in the LFP. Power in the slow oscillation band **rises** sharply at LOC and **maintains** this increase throughout the post-LOC period. Neuronal spiking **becomes** coupled to the local slow oscillation within seconds of LOC: Spiking **occurs** only in short intervals of activity that are interspersed with suppression lasting hundreds of milliseconds, periodically interrupting information processing. These periods in which activity may occur **are** not simultaneous across the brain, implying that information transfer between distant (>2 cm) cortical networks is impaired. Cortical networks therefore **are fragmented** both temporally and spatially, disrupting both local and long-range communication. However, small-scale (<4 mm) functional connectivity measures **remain similar** to the conscious state, and neuronal spike rates can recover to baseline levels after LOC despite continued unresponsiveness. **This result demonstrates** that short periods of normal spike dynamics still can occur during unconsciousness. **We conclude that** the slow oscillation **is** a fundamental component of propofol-induced unconsciousness, marking a functional isolation of cortical regions while significant connectivity **is** preserved within local networks.

1.3.2 练习2a：识别模板组件

这是来自不同领域的研究论文的两个完整的引言。通读它们并用**粗体**字或词组来识别模板组件。参考文献以**粗体**显示以引起你的注意，并且在大多数情况下，主动词为**粗体**以引起你对动词时态的注意。

1 Organic vapor phase deposition: a new method for the growth of organic thin films with large optical non-linearities[b]

Introduction

There is considerable interest in organic materials with large second-order hyperpolarizabilities for use in non-linear optical (NLO) devices such as modulators and frequency doublers **[1]**. To achieve a high figure of merit for such NLP devices **requires** a material with a non-centrosymmetric bulk structure and low dielectric constant.

To this end, NLP-active chromophores **are traditionally incorporated** into a polymer matrix and electrically poled to achieve the necessary bulk symmetry. **However**, such materials **are limited by** their **low** glass transition temperatures and **poor** stabilities at elevated temperature.

Recently, single crystals of organic and organometallic salts **[2–4] have been shown to** possess extremely large second-order ($x^{(2)}$) NLP effects leading to a high second

harmonic generation (SHG) efficiency. The naturally non-centrosymmetric crystal structures of these compounds **obviates** the need for external poling. Furthermore, these salts **have** a high optical damage threshold and sufficient stability with respect to temperature to withstand many conventional semiconductor fabrication processes. In particular, highly pure single crystals of the salt, 4'-dimethylamino-N-methyl-4-stilbazolium tosylate (DAST) **[2], have been shown to** have a value of $x^{(2)}$ at least 10^3 times greater than that of urea due to dipole alignment of the cation and anion constituents of the DAST structure. To illustrate this alignment, the DAST bulk crystal structure is shown in the inset of Fig. 1.

For many applications, such as waveguide devices, **it is desirable to** grow NLO materials into optical quality thin films. **Although** thermal evaporation in a high vacuum environment **has been used** to grow thin films of many organic **[5–7]** and inorganic materials, the technique **is not always applicable** to highly polar molecules **[8]** or organic salts. For example, when heated in vacuum, DAST **decomposes** before vaporization. Although in situ reactions of multicomponent organic molecules to synthesize polymer films **previously has been demonstrated** using vacuum techniques as physical vapour deposition or vapour deposition polymerization **[9]**, attempts in our own laboratory at double-source co-evaporation of DAST neutral precursors 4'-dimethylamino-4-stilbazole (DAS) and methyl p-toluenesulfonate (Methyltosylate, MT) to form DAST **have been unsuccessful,**

due in part to the radically different vapour pressures of DAS and MT, which leads to highly non-stoichiometric growth.

In contrast, atmospheric or low pressure (e.g. milliTorr) vapour phase epitaxy (VPE) **has been used** to grow epitaxial thin films of many III–V compound semiconductors, such as InP and GaAs, where there is a large difference in the vapour pressures of the group III and group V atomic constituents **[10]. This method was recently extended** to allow the growth of III–V and II–VI semiconductors from volatile organic precursors **[11]. Here**, a high vapour pressure compound (typically a metal halide or a metallorganic) of each respective metal **is** carried independently, via a carrier gas, to a high temperature reaction zone. In this zone, the compounds **are** deposited onto a heated substrate where they thermally **decompose** and react to yield the desired III–V compound. The excess reactants and reaction products **are** then exhausted from the system via a scrubber.

In this paper we apply the techniques of VPE to grow films of DAST by the reaction of two volatile organic materials in a hot-wall, atmospheric pressure reactor. By nuclear magnetic resonance (NMR) analysis, **we find** that the stoichiometry of polycrystalline DAST films is >95% pure (limited by instrumental sensitivity). **Using** X-ray diffraction and other analytical techniques, **we observe** a significant dependence of film quality, such as ordering and crystallite size, on the substrate composition and other deposition conditions used for growth, **suggesting that** it may be possible to generate optical quality thin films of DAST and

similar organic salts and compounds by OVPD using suitable substrates. **To our knowledge, this is the first** demonstration of the deposition of ordered thin films of a highly non-linear optically active organic salt using atmospheric vapour phase techniques.

2 Identification and characterisation of the early differentiating cells in neural differentiation of human embryonic stem cells[c]

Introduction

The developmental processes of **many** organs and tissues in an embryo **originate** from the pluripotent cells of the inner cell mass (ICM) in the blastocyst. As development proceeds, these cells gradually **acquire** specialized traits, becoming committed to specific fates and losing their potential to differentiate into other cell types. For example, the development of the central nervous system **is initiated** following gastrulation by the induction of the neuroectoderm, a process by which embryonic cells **acquire** a neural fate to form a single layer of neuroepithelial cells **[1]**. These cells subsequently **give rise to** neural stem and progenitor cells, which **undergo** further differentiation to neurons and glia **[2]**. This multi-step cell fate determination that occurs during embryonic neurogenesis **is** delicately orchestrated by many signalling pathways and transcription factors.

Although considerable efforts have been focused on ascertaining the emergence of these earliest potential neural cells and the regulatory mechanisms that govern the process of neural induction, they have **yet to be fully defined**. This is largely **due to the lack of** adequate tissues from the early developmental stages.

Human embryonic stem cells (hESCs) derived from the ICM of blastocysts **are** capable of self-renewal in culture indefinitely and meanwhile **retain** the developmental pluripotency of the embryonic founder cells, having the potential to differentiate into all the cells and tissues in a human body **[3]**. Therefore, they **provide** not only a potential source of specialized cells for regenerative therapies but also a **valuable** in vitro model to study early human development, particularly as the direct study of early human embryo development **is severely hampered by the inability** to obtain adequate amounts of tissues at all developmental stages. Although differentiation of ESCs may not fully recapitulate the development of the embryo, increasing **evidence demonstrates that** their lineage-specific differentiation nonetheless reflects the developmental progression of that cell type in vivo **[4–7]**. Therefore, the use of hESCs to investigate early human embryo development **may provide valuable insights** into early developmental processes, including neural induction.

The Oct4 transcription factor **plays** an essential role in the maintenance of pluripotency and self-renewal of ESCs **[8, 9]** and **is** also a critical reprogramming factor **[10]**. In

mouse, Oct4 **is** initially expressed in all the blastomeres of the morula, with its expression becoming successively restricted to the ICM of the blastocyst. After gastrulation, the expression of Oct4 **is** concentrated in the primitive ectoderm and **persists** through E7.5 in unsegmented areas, but **is** downregulated as development continues. By E9.5, its expression **is** limited to primordial germ cells **[11]**. In humans, Oct4 expression **remained** at stage 9 post-implantation embryos **[12]**. In the absence of Oct4, embryos **are** unable to form pluripotent ICM and **fail** to produce any other lineages, except for extraembryonic trophoblasts **[13]**. Correspondingly, forced downregulation of Oct4 in mouse and human ESCs **results** in their differentiation into extraembryonic lineages **[8, 14, 15]**. Taken together, **this implies that** Oct4 **plays** a significant role in embryogenesis and early lineage differentiation.

We have previously shown that hESCs can be efficiently differentiated to neural progenitors by the inhibition of BMP **[16]**. During the initial differentiation, expression of Oct4 **remains** detectable for at least one week until the formation of neuroepithelial cells after week 2. **Little is known about** these early differentiating cells and **it is not clear whether** this initial Oct4 expressing population differ from undifferentiated hESCs. **In this study, we** carefully **identified and isolated** this initial differentiating cell population and **demonstrated** that these cells are distinct from undifferentiated hESCs and committed neural progenitor cells (NPCs), exhibiting intermediate features between the two. The identification of

these early neural differentiating cells will provide a **valuable** cell source which **can be used** to elucidate the molecular mechanisms that regulate neuroectoderm development.

1.3.3 练习2b：识别模板组件

以下是另外六篇研究论文的完整引言。在有助于识别模板组成部分的单词和短语下画线，然后通过后面**练习2b的答案**检查你的练习。

1 The oxidative corrosion of carbide inclusions at the surface of uranium metal during exposure to water vapour[d]

Introduction

The interaction between metallic uranium surfaces and water vapour is considered to be most important in regard to the environmental corrosion of the metal. Numerous studies have examined the initial stages of these interactions [1–7]. However, there have been discrepancies in the published data describing kinetic laws, pressure dependence of the reaction rate constant and activation energies. The precise mechanism for uranium corrosion is not entirely clear with different mechanisms proposed arising from the results of the undertaking studies [1–7]. Existing discrepancies in the published data may, in part, be related to differences in the

provenance and purity of the metal used by different groups. The reactivity of impurity species such as carbide, may have affected recorded data. This work aims to provide data for an improved understanding of the role of impurity phases in the uranium–water reaction, samples of uranium containing 600 ppm carbon were analysed during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM). Samples were analysed using secondary ion mass spectrometry (SIMS), focused ion beam (FIB) imaging and sectioning and transmission electron microscopy (TEM) with X-ray diffraction (micro-XRD).

2 A GPU-tailored approach for training kernelized SVMs[c]

Introduction

Support Vector Machines (SVMs) are among the most popular general purpose learning methods in use today. SVM learning amounts to learning a linear predictor, with regularization (corresponding to a "large margin") ensuring good generalization even in very high dimensions. This predictor need not be linear in the input representation: it is possible to learn a linear predictor insome extremely high dimensional space specified implicitly through a kernel function. SVMs were originally suggested in the context of binary classification, but more recently variants following the

same principles have also been developed and successfully applied to more complex prediction tasks such as multiclass classification and prediction of structured outputs such as sequences.

Training an SVM amounts to solving a quadratic programming problem (see Section 2). Although general-purpose quadratic programming solvers can only handle fairly small SVM instances, much effort has been made in the past two decades to design special purpose solvers that can handle large-scale SVM instances. This effort resulted in widely-used packages that can solve both "linear" SVMs (i.e. where the prediction is linear in the input representation) and "kernelized" SVMs (where a non-linear kernel defines the linear prediction space). For linear SVMs, stochastic methods such as PEGASOS [13] and Stochastic Dual Coordinate Ascent [8] have recently been established as being effective at solving extremely large SVM instances, typically in less time than that which is required to read the data into memory. For kernel SVMs, most leading solvers are based on decomposing the dual optimization problem into small subproblems [11, 9, 4, 1, and see also Section 3]. Such approaches can indeed handle fairly large problems, provided that the data fits in memory, but it is not uncommon for training to require many hours or days, even using state-of-the-art optimizers. There is therefore still a strong need for faster training of kernel SVMs.

One attractive possibility for enabling faster SVM training is to leverage the power of Graphical Processing

Units (GPUs). GPUs are highly parallel, structured, computational engines and are now available relatively inexpensively and are found in many modern computers. In this paper we discuss how SVM training can be efficiently implemented on a GPU, and present such an implementation for both binary and multiclass SVMs.

Several authors have recently proposed using GPUs for kernelized SVM training [3, 2] and related problems [6]. These previous approaches, however, primarily focused on pointing out the advantages of implementing standard algorithms on graphics hardware, typically using GPU matrix-multiplication libraries, and not on how these algorithms can be modified to better take advantage of the GPU architecture. We study various algorithmic choices for SVM training in the context of GPUs, discuss how the optimal choices and algorithms on a GPU are different than those for a serial implementation, and arrive at an implementation specifically designed for graphics hardware. As with many previous approaches, we assume that the dataset fits in memory, and focus mostly on the Gaussian kernel, although our implementation can handle any kernel function which can be written in the form $K(x, y) = f(||x||, ||y||, \langle x, y \rangle)$ (see Section 5.2), and our ideas apply even more broadly to any kernel which is an aggregation of element-wise operations.

One particularly significant drawback of other GPU SVM solvers is their lack of support for sparse datasets. On the CPU, taking advantage of sparsity is a simple matter,

and sparse datasets are encountered frequently enough that many widely-used SVM solvers treat all input vectors as sparse, by default [9, 4, 1]. On the GPU, however, maximum performance is only achieved if memory accesses follow certain fairly restrictive patterns, which are difficult to ensure with sparse data. In contrast to other GPU SVM solvers, our implementation does take advantage of sparsity in the training set through a novel "sparsity clustering" approach (Section 5.3).

Overall, our implementation is orders of magnitudes faster than existing CPU implementations, and several times faster on sparse datasets than prior GPU implementations of SVM training.

3 Signals from the surface modulate differentiation of human pluripotent stem cells through glycosaminoglycans and integrins[f]

Introduction

Human pluripotent stem (hPS) cells (embryonic stem cells and induced pluripotent stem cells) are promising sources of specialized cells because of two intrinsic properties: they can self-renew indefinitely and differentiate into diverse cell types (1). To selectively guide hPS cell differentiation, precise control cellular microenvironment is needed (2, 3). Considerable effort has been devoted to identify soluble factors that promote differentiation. Additionally, there

is interest in developing matrices to support cells during differentiation (4), yet much less is known about how signals from the matrix influence cell fate.

Matrigel is the most widely used substratum for hPS cell propagation (5) and differentiation (6–12); it is derived from mouse sarcoma cells, and although its principal components are laminin, collagen, and entactin, Matrigel consists of up to 1,800 different proteins — including encapsulated growth factors — whose levels vary significantly from batch to batch (13). Accordingly, the contributions of Matrigel-delivered signals to specific phenotypic outcomes — self-renewal or differentiation — are difficult to characterize or control. Elucidating the effects of insoluble signals requires defined substrata. Such defined substrata have been developed for the longterm culture of undifferentiated hPS cells in defined media. A number of surfaces have been described including recombinant proteins (14–16), fully synthetic polymers (17–19), and peptide-modified surfaces (20–23). Surfaces that present bioactive peptides have the advantage that they can be programmed to interact with specific cell-surface macromolecules such as the glycosaminoglycans (GAGs) and integrins.

Defined surfaces can be exploited for differentiation via two distinct mechanisms. First, surfaces can be tailored to present ligands that promote adhesion of specific cell populations during differentiation (24). Second, surfaces can be devised to specifically activate (or mitigate) signal transduction pathways. Here, we use a modular approach

to exploit the aforementioned modes to control hPS cell differentiation to each of the primary germ layers. Our results reveal the advantages of defined surfaces for decoding the influence of insoluble signals on cell fate.

4 Evaluation of the damage detection capability of a sparse-array guided-wave SHM system applied to a complex structure under varying thermal conditions[g]

Introduction

Considerable effort has been expended on the development of structural health monitoring (SHM) systems [1–7], because these are widely regarded as capable of significantly reducing inspection costs of safety-critical structures in industries such as aerospace, nuclear, and oil and gas, among others. Successful SHM systems can be considered as those which combine good sensitivity to defects, preferably with the capability of localization and identification, with a low sensor density.

Nondestructive evaluation (NDE) systems based on ultrasonic guided waves generated by deployable arrays of transducers have been applied to the inspection of pipeline, rail, and relatively simple platelike structures [8–14]. These systems take advantage of the fact that certain guided-wave modes are able to propagate through several meters of the structure, often across various structural features, from a single excitation point. Recently, focus has shifted to

the development of SHM systems based on guided waves, where a small number of permanently attached transduction units would allow good coverage and real-time, on-demand inspection of complex structures [15–18].

So far, effective commercial applications of systems based on guided waves [9, 10, 13, 14] have been achieved only when processing of recorded time-traces allows straightforward identification of individual reflections, which can then be attributed either to benign structural features (e.g., stiffeners, welds, and supports) or to defects. Simple pulse-echo or pitch-catch signals from complex undamaged structures will usually consist of a large number of interfering reflections that cannot be identified; these unwanted reflections are a source of coherent noise, which is highly detrimental because it can mask reflections from defects in the structure. In these cases, it is necessary to use a benchmark or baseline signal [15, 16, 18], taken at initial stages of operation of the structure, when the possibility of the presence of defects due to fabrication processes has been eliminated by inspection through conventional NDE techniques (NDT). Because the baseline contains information on the interaction of the reflections from benign structural features in the absence of critical defects, subtraction of a current signal from such a baseline will eliminate the coherent noise to a certain degree; if cancellation of coherent noise is sufficiently good, the remaining level of residual will contain only changes in the signals caused by features that were not present initially,

such as defects [18]. The quality of the baseline subtraction therefore controls the sensitivity of the system to defects.

It is well known that environmental effects such as stress [19], ambient temperature variations [20], and liquid loading [21] affect the velocity of guided waves; this modifies the time-traces and leads to high levels of residual signal if a single baseline, taken under different conditions, is used. Of these effects, temperature variations are the most commonly encountered, and several authors have developed compensation techniques for temperature changes. These are usually based either on synthetically restoring the signal to the amplitude and phase it had at the temperature at which a reference signal was taken [16, 18, 22] or on a large and detailed look-up database of signals representing environmental conditions commonly faced by the structure [23], [24]; in this work, these methods are called optimal stretch and optimal baseline subtraction, respectively. The advantages and limitations of each of these techniques have been evaluated by different authors, and it has been shown that combining these 2 methods results in a promising temperature compensation strategy [18, 25, 26].

This paper demonstrates results obtained when a sparse-array guided-wave SHM system was applied to a shipping container panel subjected to uncontrolled temperature variations in a nonlaboratory environment. Initially, the suitability of the $A0$ and $S0$ modes for this particular structure was evaluated by studying transmission across the corrugations of the panel.

Baselines were acquired from the sparse array over several weeks and the stability of the temperature compensation and baseline subtraction techniques was assessed. Defects of different diameter were machined in the structure and the detection capability of the system was verified.

5 Silica encapsulated heterostructure catalyst of Pt nanoclusters on hematite nanocubes: synthesis and reactivity[h]

Introduction

Metal nanoparticles deposited on specific metal-oxide supports have shown a unique activity as catalysts for heterogeneous reactions.[1–4] Unfortunately, at high reaction temperatures these metal/metal-oxide composite nanostructures are unstable towards sintering.[5–7] Recent work has shown that a porous shell of silica on active nanoparticulate catalysts can stabilize the nanoparticles at high temperatures while maintaining the accessibility of the reactants to the catalyst active sites.[3,5,8] Encapsulating the metal/metal-oxide composite nanostructures using a silica shell, however, is much more challenging than for individual nanoparticles because a strong metal–support interaction is required to avoid detachment during the coating process. The deposition of metal nanoparticles on oxide support is critical for constructing such composite structures.

Hematite (α-Fe_2O_3) has a variety of applications

including pigments, gas sensors, and catalysts.[9–11] It is a highly desirable material due to its abundance, low cost, stability, safety, and its resistance to corrosion. Several high surface area nanostructures of hematite such as spherical particles,[12,13] nanorods,[14] nanocubes,[15] nanotubes,[16] and mesoporous[17] structures have been synthesized using solvothermal,[15] flame pyrolysis,[18] solution combustion,[13] electrodeposition,[16] and nanocasting methods.[17] Hematite nanostructures have been investigated as catalysts for CO and CH_4 oxidation reactions,[19] and their properties have been shown to depend upon specific crystal planes.[20,21] Halocarbons, including CCl_4, have been shown to undergo destructive adsorption on the Fe-rich (001) plane of hematite.[20,22] The adsorption of arsenate has also been studied on (012), which is known to be the most energetically stable plane of hematite.[23] The (110) plane is the next most stable following the (001) and (012) faces.[24] Relatively little is known about the reactivity of the (110) surface except the fact that it is active for CO oxidation.[15] A high CO conversion was demonstrated at 230°C over this surface.

Nanoparticles of Pd alloys,[25] Pt,[25,26] and Au[27] have shown particularly high catalytic activity when supported on hematite. High conversion and selectivity for hydrogenation reactions have been observed for Pt nanoparticles supported on γ-Fe_2O_3.[28] The deposition of metal nanoclusters on oxide surfaces has been reported using colloid deposition[26] and seed catalyzed reduction.[5,27] Both methods require pre-formed metal nanoparticles as a source.

Silica shell coatings on either metal or metal-oxide nanoparticles have been reported using sol–gel,[5,29] reverse micelles,[3] aerosol pyrolysis,[30] and atomic layer deposition.[7] Higher thermostability has been observed for the coated nanoparticles presumably due to the stability of the coating providing effective resistance to aggregation of the less stable core particles.[31] The sol–gel method is relatively inexpensive and extendable to commercial scales, and to our knowledge there has been no report of this method used to stabilize metal nanoclusters on metal oxides support nanoparticles.

In this paper the synthesis and characterization of silica encapsulated Pt/α-Fe_2O_3 heterostructures are reported. The Pt nanoparticles[32,33] were ideally deposited by photoreduction from salts on hematite nanocubes and we compared this route with hydrothermal reduction of K_2PtCl_6 in glucose solution. The following questions are addressed: (a) can Pt nanoparticles be deposited on the surface of the nanocubes and the entire heterostructure encapsulated in silica? (b) Is the core/shell heterostructure catalytically active? (c) Is the composite structure stable at the high temperatures required for catalytic reaction and regeneration?

6 Spatially resolved triboemission measurements[i]

Introduction

Triboemission is the collective term for friction-stimulated

emission of particles (electrons, ions, neutral particles and photons) that occurs under the severe conditions at the interface between sliding bodies [1]. This type of particle emission is distinct from emission due to other external stimulation, for example, by photons (photoemission [2]) or by heat (thermoemission [3]). The mechanisms that give rise to triboemission are as yet poorly understood. However, it is believed to initiate the chemical reactions that lead to the formation of surface films in boundary lubrication [4, 5]. For example, widely used zinc dialkyldithiophosphate (ZDDP) additives produce boundary films by a mechanism that cannot be driven by contact temperatures and pressure alone [6]. In addition to this, triboemission has been blamed for the degradation of computer hard drive lubricants [7] and has also been linked to flash lights during the fracture of rocks associated in earthquakes [8] and volcanic eruptions [9]. In order to understand the role of emitted particles in these applications — so that boundary lubrication can be enhanced and lubricant degradation prevented — it is necessary to study the mechanisms by which emission occurs. For this reason, triboemission has been the focus of significant research efforts for over half a century [10].

The first significant triboemission measurements were made by Nakayama *et al.* [11] during scratch tests by a diamond tip. To achieve this, they used a low-voltage-biased collector positioned close to the contact that registered bursts of the electrical current due to the emission of negatively

and positively charged particles, while photon emission was measured using a photomultiplier [12]. Their results showed that insulator materials produced significantly higher levels of emission when scratched than metals. Kim *et al.* [13] investigated the emission of electrons using channel electron multipliers (CEMs) and photons using photomultipliers, during abrasion of MgO samples under vacuum. They also measured the kinetic energy of the electrons using a retarding grid. In these studies, photon emission was attributed mainly to deformation and electrons emission to fracture. Molina *et al.* investigated the triboemission of negatively charged particles under vacuum for a wide range of materials (alumina, sapphire, silicon nitride and semiconductors) using a CEM in pulsecounting mode [14–16]. The results from Nakayama *et al.* and Molina *et al.* suggested that triboemission is related to material hardness [14, 17]. Emission is also correlated strongly with the wear of the surface, highlighted by the observation that repeated scratches along a single track result in a decrease of emission with time, due to decreasing wear [14].

From these investigations, it is generally concluded that triboemission of electrons is related to surface fracture during sliding contact. More specifically, Dickinson *et al.* [18] put forward a charge separation model, as shown in Fig. 1. Here, an imbalance of charge is created between opposite faces of a crack as it opens. Due to the small width of the crack, this electric field has a large gradient, which encourages electrons to leave the negatively charged crack face. Due to thermal

vibrations, a portion of these electrons do not reach the opposing face and are instead emitted into the surrounding environment. This model, however, does not explain why the energy of the tribo-emitted electrons can be lower than the work function of the sample [14]. To account for these low-energy electrons, Dickinson *et al.* [18] suggested that the work function of the surface could be locally decreased by small defects in the highly strained areas.

Since triboemission measurements typically contain no spatial information, researchers have tried to gain insights into the emission mechanisms by analysing the temporal characteristics of the recorded signals. For example, Molina *et al.* [15] showed that emission patterns are not random and proposed stochastic and chaos-deterministic approaches [16] to characterize their evolution with time. Recently, Nakayama *et al.* [19, 20] and Matta *et al.* [21] have made significant advances in infrared and ultraviolet imaging of the triboplasma, generated at the rear of sliding contacts by gas discharge under ambient conditions. The research described in the current paper differs from these studies, since we are concerned with electron emission mechanisms before interactions with surround gas take place.

One of the main limitations of electron triboemission experiments to date is that they provide only ensemble average values, giving no information relating to spatial location or direction. This shortcoming is, in part, responsible for our lack of understanding of triboemission processes,

since it prevents emission measurements from being directly correlated with other surface analyses, such as SEM and AFM. To address this issue, we present experiments using a MCP, in which spatial maps of electron emission are obtained (see Fig. 2). The MCP — which is effectively a dense array of electron multipliers — is located within a vacuum chamber above a stationary diamond stylus that is loaded against a rotating aluminium disc covered with a 5-μm oxide layer (a brittle material on a soft substrate is likely to form cracks and thus induce emission). In this way, bursts of electrons can be visualized, revealing the pattern, direction and temporal characteristics of emission.

练习3　分析和建立你的目标研究论文引言部分的模板。完成此操作后，你将会以不同的方式进行阅读：除了阅读内容以外，你还将了解该内容的表达方式。了解段落和句子是如何开始的，它们是如何连接到一起的。你将开始看到整合信息的支架，该支架将成为一种终身受用的写作阅读工具，每当你阅读综述论文、学术论文、会议摘要或其他文章时，它都会自然而然地更新。

练习2b的答案

1 The oxidative corrosion of carbide inclusions at the surface of uranium metal during exposure to water vapour[d]

Introduction

The interaction between metallic uranium surfaces and water vapour is considered to be most important in regard to the environmental corrosion of the metal. Numerous studies have examined the initial stages of these interactions [1–7]. However, there have been discrepancies in the published data describing kinetic laws, pressure dependence of the reaction rate constant and activation energies. The precise mechanism for uranium corrosion is not entirely clear with different mechanisms proposed arising from the results of the undertaking studies [1–7]. Existing discrepancies in the published data may, in part, be related to differences in the provenance and purity of the metal used by different groups. The reactivity of impurity species such as carbide, may have affected recorded data. This work aims to provide data for an improved understanding of the role of impurity phases in the uranium–water reaction, samples of uranium containing 600 ppm carbon were analysed during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM). Samples were analysed using secondary ion mass spectrometry (SIMS), focused ion beam

(FIB) imaging and sectioning and transmission electron microscopy (TEM) with X-ray diffraction (micro-XRD).

2 A GPU-tailored approach for training kernelized SVMs[c]

Introduction

Support Vector Machines (SVMs) are among the most popular general purpose learning methods in use today. SVM learning amounts to learning a linear predictor, with regularization (corresponding to a "large margin") ensuring good generalization even in very high dimensions. This predictor need not be linear in the input representation: it is possible to learn a linear predictor in some extremely high dimensional space specified implicitly through a kernel function. SVMs were originally suggested in the context of binary classification, but more recently variants following the same principles have also been developed and successfully applied to more complex prediction tasks such as multiclass classification and prediction of structured outputs such as sequences.

Training an SVM amounts to solving a quadratic programming problem (see Section 2). Although general-purpose quadratic programming solvers can only handle fairly small SVM instances, much effort has been made in the past two decades to design special purpose solvers that

can handle large-scale SVM instances. This effort resulted in widely-used packages that can solve both "linear" SVMs (i.e. where the prediction is linear in the input representation) and "kernelized" SVMs (where a non-linear kernel defines the linear prediction space). For linear SVMs, stochastic methods such as PEGASOS [13] and Stochastic Dual Coordinate Ascent [8] have recently been established as being effective at solving extremely large SVM instances, typically in less time than that which is required to read the data into memory. For kernel SVMs, most leading solvers are based on decomposing the dual optimization problem into small subproblems [11, 9, 4, 1, and see also Section 3]. Such approaches can indeed handle fairly large problems, provided that the data fits in memory, but it is not uncommon for training to require many hours or days, even using state-of-the-art optimizers. There is therefore still a strong need for faster training of kernel SVMs.

One attractive possibility for enabling faster SVM training is to leverage the power of Graphical Processing Units (GPUs). GPUs are highly parallel, structured, computational engines and are now available relatively inexpensively and are found in many modern computers. In this paper we discuss how SVM training can be efficiently implemented on a GPU, and present such an implementation for both binary and multiclass SVMs.

Several authors have recently proposed using GPUs for kernelized SVM training [3, 2] and related problems [6]. These previous approaches, however, primarily focused

on pointing out the advantages of implementing standard algorithms on graphics hardware, typically using GPU matrix-multiplication libraries, and not on how these algorithms can be modified to better take advantage of the GPU architecture. We study various algorithmic choices for SVM training in the context of GPUs, discuss how the optimal choices and algorithms on a GPU are different than those for a serial implementation, and arrive at an implementation specifically designed for graphics hardware. As with many previous approaches, we assume that the dataset fits in memory, and focus mostly on the Gaussian kernel, although our implementation can handle any kernel function which can be written in the form $K(x, y) = f(\|x\|, \|y\|, \langle x, y \rangle)$ (see Section 5.2), and our ideas apply even more broadly to any kernel which is an aggregation of element-wise operations.

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Overall, our implementation is orders of magnitudes

faster than existing CPU implementations, and several times faster on sparse datasets than prior GPU implementations of SVM training.

3 Signals from the surface modulate differentiation of human pluripotent stem cells through glycosaminoglycans and integrins[f]

Introduction

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4 Evaluation of the damage detection capability of a sparse-array guided-wave SHM system applied to a complex structure under varying thermal conditions[g]

Introduction

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5 Silica encapsulated heterostructure catalyst of Pt nanoclusters on hematite nanocubes: synthesis and reactivity[h]

Introduction

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6 Spatially resolved triboemission measurements[i]

Introduction

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From these investigations, it is generally concluded that triboemission of electrons is related to surface fracture during sliding contact. More specifically, Dickinson *et al.* [18] put forward a charge separation model, as shown in Fig. 1. Here, an imbalance of charge is created between opposite faces of a crack as it opens. Due to the small width of the crack, this electric field has a large gradient, which encourages electrons to leave the negatively charged crack face. Due to thermal vibrations, a portion of these electrons do not reach the opposing face and are instead emitted into the surrounding environment. This model, however, does not explain why the energy of the tribo-emitted electrons can be lower than the

work function of the sample [14]. To account for these low-energy electrons, Dickinson *et al.* [18] suggested that the work function of the surface could be locally decreased by small defects in the highly strained areas.

Since triboemission measurements typically contain no spatial information, researchers have tried to gain insights into the emission mechanisms by analysing the temporal characteristics of the recorded signals. For example, Molina *et al.* [15] showed that emission patterns are not random and proposed stochastic and chaos-deterministic approaches [16] to characterize their evolution with time. Recently, Nakayama *et al.* [19, 20] and Matta *et al.* [21] have made significant advances in infrared and ultraviolet imaging of the triboplasma, generated at the rear of sliding contacts by gas discharge under ambient conditions. The research described in the current paper differs from these studies, since we are concerned with electron emission mechanisms before interactions with surround gas take place.

One of the main limitations of electron triboemission experiments to date is that they provide only ensemble average values, giving no information relating to spatial location or direction. This shortcoming is, in part, responsible for our lack of understanding of triboemission processes, since it prevents emission measurements from being directly correlated with other surface analyses, such as SEM and AFM. To address this issue, we present experiments using a MCP, in which spatial maps of electron emission are obtained

(see Fig. 2). The MCP — which is effectively a dense array of electron multipliers — is located within a vacuum chamber above a stationary diamond stylus that is loaded against a rotating aluminium disc covered with a 5-μm oxide layer (a brittle material on a soft substrate is likely to form cracks and thus induce emission). In this way, bursts of electrons can be visualized, revealing the pattern, direction and temporal characteristics of emission.

1.4 有用的词汇和短语

1.4.1 语言任务

练习4 仔细阅读本单元的引言和你的目标研究论文的引言。给所有可以用在下面四个方面的单词或短语画线或做标记，然后将你的建议与第1.4.2节中列出的单词和短语进行比较。例如：

1. 确立研究的重要性

寻找如 widespread and much research in recent years 之类的单词和短语。

2. 已有和/或目前的研究和贡献

你不能总是一成不变地使用老套的词句（如they did/showed/found），有时你需要更加具体。因此寻找描述具体执行

了哪些操作的动词(如 calculated、monitored、identified)。

3. 关于空白/问题/预测

寻找准确的方式来表达已有的和/或当前的研究尚不完整,或者没有解决你的论文所涉及的问题(如 inefficient、unclear、few studies have focused on…)。

4. 当前的论文

寻找对当前论文的目标、策略或优势的描述(如we propose、our approach、successful)。

1.4.2 引言使用的语言

本节列出了通过对2500多篇不同学科已发表研究论文的分析所得到的用于引言的单词和短语。这个列表仅包含经常出现的单词和短语,因此被作者和编辑视为常规且可以接受的。

该列表还会有助于你保持流畅向前的写作状态。在每个列表的下面都有一些示例,说明单词和短语在句子中的使用方式,因此,当你感到困惑并且想不出该写什么的时候,查看句子示例以及列表。

1. 确立研究的重要性(通常在第一句话呈现)

(an) advantage	attracted (much) attention
(an) attractive approach	benefit/beneficial

(a) central problem	common/ly
(a) challenging area	cost-effective
(a) considerable number	during the past (two) decades
(a) crucial issue	emerging
(a) current challenge	extensively studied
(a) dramatic increase	for many years
(an) essential element	frequent/ly
(a) focus on	great potential
(a) fundamental issue	importance/important
(a) global concern	major
(a) growth in popularity	many/most
(an) increasing number	multidisciplinary interest
(an) interesting aspect	much study in recent years
(a) key technique	now
(a) leading cause (of)	numerous investigations
(a) number of	of growing/great interest
(a) popular method	often
(a) powerful tool/metho	play a key/major role (in)
(a) primary cause (of)	principle
(a) (wide) range (of)	recent/ly
(a) rapid development	relevant
(a) remarkable variety	several
(a) significant increase	today
(a) striking feature	typical
(a) traditional technique	valuable

(a) useful method	well-documented
(a) variety of	well-known
(a) vital aspect	wide/ly
(a) worthwhile study	widespread
	worldwide

下面是来自研究论文引言的第一句话的一些例子：

- The status and function of the port has evolved **rapidly in recent years**.
- Since the discovery of the first isolated graphene layer, **many** chemical approaches have been developed.
- The **increasing** pressure to eliminate lead has stimulated **great interest** in the search for lead-free solders.
- Fourier transforms are **widely used** in image processing to characterise textures in images.
- The vanadium redox flow battery (VRFB) is an **emerging** energy storage technology.
- Steady-state multiphase upscaling has become **increasingly popular** because it is **fast, robust and computationally cheap**.
- The optic thalamus has been the subject of **numerous investigations**.
- Malaria affects as **many** as 216 million people annually, with 445,000 deaths occurring primarily in children under 5 years old.*
- Control of serum phosphorus levels is **a central goal in** the

management of patients with chronic renal failure.

- Heat transfer phenomena **play an important role in** welding.

*注：这些数字可能会随着时间的推移而变化，因此请考虑加入日期（例如，在2019年）。

2. 用于指代/描述先前和目前研究活动及贡献的动词

achieve	create	focus on	produce
address	deal with	generate	propose
analyse	define	identify	prove
apply	demonstrate	imply	provide
argue	describe	improve	put forward
assess	design	interpret	recognise
assume	detect	introduce	report
attempt	determine	investigate	resolve
calculate	develop	measure	reveal
categorise	discover	mention	review
carry out	discuss	model	revise
challenge	enhance	modify	show
claim	establish	monitor	solve
clarify	estimate	note	state
collect	evaluate	observe	study
compare	examine	obtain	suggest
conclude	explain	perform	support
conduct	explore	point out	test

confirm	extend	predict	use
consider	find	present	verify

注：当你想表明你的研究内容或论文内容时，你可以在引言的任何地方使用这些动词，如在引言末尾。

下面是一些应用的例子：

- The effect of keratin on the mechanical properties of copolymer systems **was demonstrated** more than two years ago.
- Minelli *et al.* **describe** the use of experts' opinions and empirical evidence.
- NDE systems **have been applied** to the inspection of pipelines.
- Several authors **have developed** compensation techniques for temperature changes.
- The advantages and limitations of each of these techniques **have been evaluated** by different authors.
- Kim *et al.* **investigated** the emission of electrons using channel electron multipliers (CEMs).
- Dickinson *et al.* **put forward** a charge separation model.
- For simplicity, they only **considered** the homogeneity ranges in their model.
- They were the first to **recognise** the importance of studying the effect of genetic variation.
- Initial attempts **focused on determining** the cause of…
- The results on pair dispersion **are reported in**…

3. 已有研究的空白

第一组：问题/批评

absent	inflexible	(a) challenge
complicated	insufficient	(a) defect
controversial	misleading	(a) difficulty
costly	not able to…	(a) disadvantage
deficient	not ideal	(a) drawback
disappointing	not sufficiently…	(a) flaw
doubtful	not/no longer useful	(a) gap
expensive	of little value	(a) lack
false	over-simplistic	(a) limitation
far from (ideal)	poor	(an) obstacle
fragile	problematic	(a) problem
hard to (detect)	restricted	(a) risk
impractical	severe	(a) shortcoming
inaccurate	time-consuming	(a) weakness
inadequate	unable to	(to) be confined to
incapable (of)	undesirable	(to) fail to
incompatible	unnecessary	(to) fall short of
inconsistent	unrealistic	(to) ignore
inconvenient	unsatisfactory	(to) lag behind
incorrect	unsuccessful	(to) miscalculate
ineffective	unsuitable	(to) misjudge
inefficient	unsupported	(to) misunderstand

inferior		(to) neglect (to) overlook (to) suffer (from)

注：这些词汇通常会通过如however、although、while、nevertheless 和 despite 等单词来进行示意。

第二组：研究机会

(an) alternative	more work is needed
(the) next step	not addressed
(to) demand clarification	not dealt with
(to) need to re-examine	not studied
(to) raise the question	not well understood
(to) remain controversial	suggest/s that…
(to) remain unstudied	there is an urgent need…
(to) require (clarification)	to the best of our knowledge
few studies have…	unanswered
ideal candidate	unclear
incomplete	unexamined
it is/seems possible that…	unproven
little evidence is available	unsolved
little work has been done	

注：研究机会通常通过如may/might/could/would 等情态动词来传达。

下面是一些应用的例子：

- The fact that these galaxies are so massive may make them **ideal candidates** for protocluster searches.
- **Little attention has been paid** to the selection of an alternative biocompatible material.
- How propofol acts on neural circuits to produce unconsciousness **remains unclear**.
- The main **drawback** of this approach is that it is **unable** to provide quantitative information.
- **Little is known** about these early differentiating cells and **it is not clear** whether this initial Oct4 expressing population differ from undifferentiated hESCs.
- One of the **limitations** of electron triboemission experiments to date is that they provide **only** ensemble average values.
- This **shortcoming** is responsible for our **lack of understanding** of triboemission processes.
- **However**, the search for a stable oral prostaglandin preparation has been **largely unsuccessful**.
- The high absorbance makes this **an impractical option**.
- **An alternative approach** is necessary.
- Determining the function of these proteins **remains a challenge**.
- **Although** there is general agreement regarding the timing of sea-level events, their amplitude **remains controversial**.
- These can be **time-consuming** and are often **technically difficult** to perform.

4.当前的工作

(to) attempt	(is) organised as follows:	**'happy' words** ☺
(to) compare	(is) set out as follows:	able to
(to) concentrate (on)	(our) approach	accurate
(to) describe	(the) present work	advantage
(to) determine	(this) paper	effective
(to) develop	(this) project	efficient
(to) discuss	(this) report	excellent
(to) enable	(this) section	fully
(to) enhance	(this) study	innovation
(to) evaluate	(this) work	new
(to) facilitate	aim	novel
(to) focus on	goal	potential
(to) identify	intention	powerful
(to) improve	objective	practical
(to) investigate	purpose	promising
(to) offer		relevant
(to) outline		robust
(to) predict		simple
(to) present		straightforward
(to) propose		successful
(to) provide		superior
(to) report		unique
(to) reveal		valuable
(to) solve		
(to) succeed		

下面是一些应用的例子：

- **Our study focuses on**…
- To address this question, **we investigated**…
- Section V **provides several relevant** conclusions obtained from **the study reported in this paper**.
- **The main objective of this study** was **to describe and examine**…
- **In this paper we present** a **robust** method for…
- **The proposed method** combines GWAS data with information from bioinformatics databases in a **coherent and reproducible** way.
- **In this study**, we **carefully identified and isolated**…
- **New** correlations were developed with **excellent** results.
- **We show that** use of glass reactors can be **effective**.
- **To our knowledge, this is the first** demonstration of the…
- **This paper introduces** a scheme which **solves** these problems.
- **To address this issue, we present** experiments using an MCP.
- **Our design** is both **simple** and **accurate**, and can be **fully** integrated into…
- **This paper is organised as follows**:

1.5　语言和写作技巧

本节讨论在引言和其他地方都很重要的四个主题：

- 动词时态的选择
- 句子和信息的连接

- 被动/主动语态的用法
- 分段

1.5.1 动词时态的选择

如我们所见，大多数作者在引言中呈现已有的研究和贡献：研究“地图”。他们需要用不同的动词时态来完成此操作，通常包括过去时、一般现在时和现在完成时。要使用三个时态中的哪一个很少是根据语法规则来决定的，在大多数情况下是根据上下文的含义做出决定的。使用特定的动词时态来介绍已有的研究和贡献，这代表的是作者对该句子功能的一种选择，并非出于语法上的要求。根据你想就特定研究或贡献表达的内容，或根据你希望读者理解这些内容的方式，你可以在过去时、一般现在时和现在完成时之间进行选择。在本节中，我们将会看到如何决定这些选择。

在过去时和现在时之间进行选择

They found that the pressure **increased** as the temperature **rose**.

They found that the pressure **increases** as the temperature **rises**.

They found that the dielectric structure **possessed** a full photonic bandgap.

They found that the dielectric structure **possesses** a full photonic bandgap.

使用过去时仅仅是描述作者在研究中发现的内容，这些发现与当时的研究相关联，但是并不作为总是成立的事实。相比之下，选择现在时则反映了一种信念，即当时的发现足够强有力并且可靠，可以构成一个永久的事实。

但是请注意，正如我们在第1.2.2节中所看到的那样，许多以现在时态呈现的背景事实始于作为过去时态陈述的研究发现，并随着它们的发展和接受而逐渐演变为现在时。这意味着五年前的期刊文章可能会使用不再合适于现在的动词时态，而现在仍然使用过去的时态会给人一种你不了解该领域的最新动态的印象。当你撰写有关已有研究和著作的论文时，**要检查针对该特定发现或事实在当前使用的动词时态**。搜索引擎（例如Google Scholar）是对此检查的一个很好的资源，但是请确保你仅搜索最新的研究。

在过去时和现在完成时之间进行选择

They **found** that the pressure increases as the temperature rises.

They **have found** that the pressure increases as the temperature rises.

They **found** that the dielectric structure **possesses** a full photonic bandgap.

They **have found** that the dielectric structure **possesses** a full photonic bandgap.

请看下面的句子：

（a）过去时	I lived in Tokyo for five years	……但是现在我不住在那里了
（b）现在完成时	I have lived in Tokyo for five years	……**现在**我还住在那里

句(a)和句(b)之间的区别是事件的“时间”，即作者是否仍住在东京。但是，在这些句子中：

（c）过去时	I broke my glasses	……但是我还有一副/我修好它了
（d）现在完成时	I have broken my glasses	……所以我**现在**看不清楚了

两者之间的区别更加复杂，对于作为STEMM科研作者的你来说也更加重要。在(c)和(d)中，事件的“时间”不是关键的区别，(c)和(d)都有可能发生在上个月，今天早上或极短的时间之前。(d)中的事件采用现在完成时态，这是因为与(c)中的事件相比，它**与现在的情况更相关**。为什么在撰写引言时这种与当前相关性的想法是很重要的？从第1.2.1节中的引言中看这句话：

However, although the effect of keratin on the mechanical properties of copolymer systems **was demonstrated** over two years

ago[9], keratin adds considerably to the weight of the composite, and little attention **has been paid** to the selection of an alternative biocompatible material.

在这句话中，作者从过去时（**was demonstrated**）改变为现在完成时（**has been paid**），以便表达后者是当前研究中的空白。在这里选择现在完成时以表示该研究论文**从现在开始将关注合适的生物相容性材料的选择问题**（the selection of an appropriate biocompatible material）。

将此与下面的句子进行比较。在下面这个句子中，作者没有改变时态，而是继续使用过去时：

However, although the effect of keratin on the mechanical properties of copolymer systems **was demonstrated** over two years ago[9], keratin adds considerably to the weight of the composite, and little attention **was paid** to the selection of an alternative biocompatible material.

通过使用过去时，作者表示当时未引起注意，例如两年前。也许从那时起此问题已经受到关注，或者有可能此问题已经得到解决了。时态的变化总是有意义的，它们总是标志着信息功能的变化，因此请不要随意选择或更改时态。

查看你的目标文章的引言部分使用过去时和现在完成时的方式，来检查有关时态使用的信息。特别是了解这些文章在描述

已有研究及其贡献时使用一般过去时或者现在完成时的方式。

1.5.2 句子和信息的连接

随着句子的进展，读者可以更轻松地预测句子的去向。看一下以下句子：

If you're paying attention to this sentence, you can probably predict what the last word will ___.

预测该句子的最后一个单词相当容易，但是预测下一个句子的第一个单词则要困难得多。在每个句子的末尾，作者停下来思考下一步该写什么，而在这样的思考时间（可能是几秒钟，几小时甚至几天）中，两个句子之间的距离开始拉长，不利于读者理解其含义。句子之间的空白是一个危险的空间。作为作者，你的工作是尽可能缩小这片空白，使读者可以轻松地将一条信息与另一条信息联系起来。连接信息不仅对读者有益，对作者也有益，因为它会鼓励他们以合乎逻辑的方式发展相关的思想和文本。

一个句子的开头和连接到前一个句子的方式是它成功的关键，包括段落的成功和整个文本的成功。有四种方法开始一个句子，可以使之与前面的信息清楚地联系。第一种方法是**重叠**，即**在下一个句子的开头重复上一个句子的内容**。请看下面的例子：

This steady-state approach cannot distinguish between patterns that are characteristic of a deeply anesthetized brain and those that arise at the onset of **unconsciousness. Unconsciousness** can occur in tens of seconds…

… implying that information transfer between distant (>2 cm) **cortical networks** is impaired. **Cortical networks** therefore are fragmented both temporally and spatially, disrupting both...

The pattern of inflammation during an asthma attack is different from that seen **in stable asthma. In stable asthma** the total number of inflammatory cells does not increase.

…although metals such as aluminium have been used **to strengthen** PLA for industrial applications[5,6], these are not appropriate for biomedical use[7].One way **to strengthen** polymers for biomedical applications is…

Elucidating the effects of insoluble signals requires **defined substrata**. Such **defined substrata** have been developed for the long-term culture of undifferentiated hPS cells in defined media.

第二种方法是使用**替代形式**（This method、These systems）将句子连接在一起。请看下面的例子：

On the basis of **these criteria** it then describes the preparation of a biomass-derived polymer blend using chitin, a long-chain polymer found in many places throughout the natural world. **This combination** formed a novel lightweight copolymer…

The effects on macroscopic dynamics are noticeable in

the EEG, which contains several stereotyped **patterns** during maintenance of propofol general anesthesia. **These patterns** include increased delta power…

Most studies have focused on a deep **steady state** of general anesthesia and have not used a systematic behavioral measure to track the transition into unconsciousness. **This steady-state** approach cannot distinguish between…

Many researchers have suggested **ways** to reduce cost without affecting the quality of the image. **These methods** rely on data structures built during a preprocessing step.

One of the main **limitations** of electron triboemission experiments to date is that they provide only ensemble average values, giving no information relating to spatial location or direction. **This shortcoming** is, in part, responsible for our lack of understanding…

需要注意的是，当你使用替代形式时，如果你重复使用与第一次使用的完全相同的名词，对读者来说更容易。如果你将某事物描述为一种设备（a device），下次引用时应该是该设备（this device），而不是该技术（this technique）或该方法（this method）。

你可以在句子的开头用 It、They、These 或者 This 来引用，但是对于读者来说可能很难弄清这些单词所指的具体含义。请看下面的例子：

- This result was statistically significant and is in line with results in previous studies. **This** suggests that...

这里的**This**是指 “the result was statistically significant.” 还是 “the result is in line with results in previous studies.”？或者两者皆是?

- The existence of small amounts of impurities will change these properties and **they** are easily detectable.

这里**They**是指 impurities，还是small amounts of impurities，还是these properties?

第三种方法是根本不结束句子，而是用**分号**把上下文的句子连接起来。当两个句子关系非常紧密并且其中一个非常短的时候，用分号连接句子能起到很好的效果。以分号连接句子的方式感觉像是长逗号而不是句号。

The procedure for testing whether components are operationally safe usually takes many hours; this means that tests are rarely repeated.

Simple pulse-echo or pitch-catch signals from undamaged structures usually consist of interfering reflections that cannot be identified; these unwanted reflections are a source of coherent noise, which can mask defects in the structure.

Due to the small width of the crack, this electric field has a large gradient; this encourages electrons to leave the negatively charged crack face.

句子的长度是影响可读性的重要因素，因此如果你用分号连接句子，请检查句子的整体长度。第一次阅读的时候，90%的读者能理解少于20个单词的句子；只有10%的读者能理解那些单词数超过40的句子。STEMM研究论文中的句子的平均长度为23个单词。

第四种方法是使用一个信号（如therefore 或however）来传达句子的功能。如果正确使用它们会很有帮助，但是这些信号词不是简单地将句子“黏合”在一起，如果没有正确使用它们，则弊大于利。以 Moreover开头的句子告诉读者，他们正在阅读的句子的功能与上一个句子相同。换句话说，如果前一句提出了做某件事的理由，那么本句子也必须做同样的事情。如果前面的句子表达在某些方面的缺点，那么本句子必须表达其他缺点。类似地，以Therefore或Consequently开头的句子必须呈现或描述与可识别的原因或理由直接相关的效果、结果或产物。这些因果关系应该对读者来说是清楚明白的或明确的，而不是主要存在于作者的脑海中。如果不是很明显，则可能在使用Therefore 之前需要更多的信息。

以下是根据信号词的功能安排的一些示例。这不是一个长的列表，因为仅包括当前STEMM写作中常见的一些例子。

表示原因

The experiment was unsuccessful ____ the measuring instruments were inaccurate.

The experiment was unsuccessful ____ the inaccuracy of the measuring instruments.

due to (the fact that)	as
on account of (the fact that)	because
in view of (the fact that)	since

as也可能表示when 的意思，所以如果有引起混淆的任何可能性，需要选择一个不同的连接词。例如，不清楚下面句子中的 **As** 是指 Because 还是When：“**As** repeated melting homogenised more material, the phase transitions became less prominent.”

since 也可能表示from that time，所以如果有引起混淆的任何可能性，需要选择一个不同的连接词。

表示结果

The measuring instruments were calibrated accurately, ___ the experiment was successful.

therefore	hence	thus
consequently	as a result	so

表示对比/差异

British students were all vegetarians, ___ Norwegian students eat meat every day.

however whereas but	on the other hand while	by contrast in contrast

- on the contrary 和 conversely 并不适合这一范畴，因为它们不仅表达差异，它们表达的是“恰好相反”这一事实，因此不能用在上面的句子里（因为素食主义者和食肉者不是对立的，他们只是不同而已）。然而你可以把它们用在如下面的句子里：“Some experiments used uncalibrated instruments and succeeded; **conversely**, other experiments used carefully calibrated instruments and failed.”
- 还要记住，while意味着 at that/at the same time，所以如果有任何混淆的可能性，请选择其他信号词，例如whereas。

表示意外

A _____ it was difficult, a solution was quickly found.

B _____ the difficulty, a solution was quickly found.

C It was difficult; ____ a solution was quickly found.

A	B	C
although even though	despite in spite of	nevertheless however

though	regardless of notwithstanding	yet but nonetheless even so

请注意，however和but除了表示意外，还可以用来表示相反/差异，因此如果你想强调意外情况，请从此列表中选择其他信号词。

表示并列/清单

We used a batch processing system because it was more effective. _____ it was significantly less expensive.

in addition moreover furthermore	also secondly (etc.) in the second place (etc.) what is more,

请注意，besides 跟上面列的各项意思差不多，但是更加有力，因此最好用在更具有说服力的语境中。

表示过渡

The drug is extremely effective and easy to administer. ____ cost, it is approximately the same price per dose as existing drugs.

with regard to	with respect to
as to	as for
regarding	turning now to

不必每个句子都以信号词开头(而且这样看起来也很公式化)。信号词是有强调性的,并且每个句子都以信号词开头会得到一个晦涩和过度强调的文本。为确保文本中的信息合乎逻辑且易于遵循,请考虑使用重复连接而不是信号词。这意味着在句子之间,尤其是在句子的开头重复单词,以确保可以将读者从一项信息小心地带到另一项信息中。

检查句子和信息是否明确地连接是一种很有价值的自我编辑工具,因为它会迫使你检查句子之间的关系。良好的连接有助于文章的流畅性。

这下面关于论文 "Rapid fragmentation of neuronal networks at the onset of propofol-induced unconsciousness" 的引言中,句子到句子的连接以**粗体**显示,重复连接以***斜体***显示。令人吃惊的是,文本中有多处连接和重复——结果是使阅读和理解变得如此容易。查看完这个示例后,请在你的目标文章的引言和本单元的其他引言中检查句子连接和重复连接的方式。

> General anesthesia is a drug-induced reversible coma commonly initiated by administering a large dose of a fast-acting *drug* to induce unconsciousness within seconds (1).

This state can be maintained as long as needed to execute surgical and many nonsurgical procedures. One of the most widely used *anesthetics* is propofol, an i.v. *drug* that enhances GABAergic inhibitory input to neurons (2–4), with effects in cortex, thalamus, brainstem, and spinal cord (5–7). **Despite** the understanding of *propofol's* molecular actions, it is not clear how *these effects* at molecular targets affect single *neurons* and larger-scale neural circuits to produce *unconsciousness*.

The effects on macroscopic dynamics are noticeable in the EEG, which contains several stereotyped **patterns** during maintenance of *propofol general anesthesia*. **These patterns** include increased delta (0.5–4 Hz) power (8, 9); increased gamma (25–40 Hz) power (9); an alpha (~10 Hz) rhythm (10–12) that is coherent across frontal cortex; and burst suppression, an alternation between bursts of high-voltage activity and periods of flat *EEG* lasting for several seconds (13, 14). **In addition**, slow oscillations (<1 Hz) have been well characterized in deeply anesthetized animals and are associated with an alternation of the neuronal membrane potential between UP (depolarized) and DOWN (hyperpolarized) states (8, 15).

Although these patterns are observed consistently, it is unclear how they are functionally related to *unconsciousness* under *general anesthesia*. Most studies have focused on a deep **steady state** of *general anesthesia* and have not used a systematic behavioral measure to track the transition

into *unconsciousness*. **This steady-state approach** cannot distinguish between *patterns* that are characteristic of a deeply anesthetized brain and those that arise at the onset of **unconsciousness. Unconsciousness** can occur in tens of seconds (4), but many neurophysiological features continue to fluctuate for minutes after induction and are highly variable between different levels of *general anesthesia* (1,16). **Therefore**, identifying the specific dynamics associated with loss of consciousness (LOC) requires an examination of the *transition into unconsciousness*, linking neurophysiology with *behavioral measures*.

In addition, the dynamic interactions between cortical areas that underlie *these EEG oscillations* are not well understood, because few studies have simultaneously recorded ensembles of *single neurons* and oscillatory dynamics from sites distributed across the brain. **Consequently**, how *propofol* acts on neural circuits to produce *unconsciousness* remains unclear. A leading hypothesis suggests that anesthetics disrupt cortical integration (17, 18). Identifying the mechanism by which *this disruption* might occur requires a better understanding of how the spatial and temporal organization of neural dynamics evolves during *induction* of *unconsciousness*.

To address this question, we investigated both *neuronal* and circuitlevel dynamics in the human brain during *induction* of *unconsciousness* with *propofol*. **We** obtained simultaneous recordings of single units, local field potentials (LFPs), and intracranial electrocorticograms (ECoG) over up to 8 cm of

cortex, enabling us to examine neural dynamics at multiple spatial scales with millisecond-scale temporal resolution. **We** used a *behavioral* task to identify within seconds the time at which patients became unresponsive to auditory stimuli, which we defined as *LOC*.

Our results reveal a set of neurophysiological features that accompany *LOC* that, together with previously reported effects (8, 9, 15), enable a multiscale account of this profound shift in brain state. **We** find that *LOC* is marked by the abrupt onset of *slow oscillations* (0.1–1 Hz) in the LFP. Power in the *slow oscillation* band rises sharply at *LOC* and maintains this increase throughout the post-*LOC* period. *Neuronal* spiking becomes coupled to the local *slow oscillation* within seconds of *LOC*: **Spiking** occurs only in short intervals of activity that are interspersed with suppression lasting hundreds of milliseconds, periodically interrupting information processing.**These periods** in which activity may occur are not simultaneous across the brain, implying that information transfer between distant (>2 cm) *cortical* networks is impaired.**Cortical networks** therefore are fragmented both temporally and spatially, *disrupting* both local and long-range communication. **However**, small-scale(<4 mm) functional connectivity measures remain similar to the conscious state, and *neuronal spike* rates can recover to baseline levels after *LOC* despite continued unresponsiveness.**This result** demonstrates that short periods of normal *spike* dynamics still can occur during *unconsciousness*.**We** conclude that the *slow*